

SURVEY

Empowering Security Operation Center With Artificial Intelligence and Machine Learning—A Systematic Literature Review

MOHAMAD KHAYAT¹, EZEDIN BARKA¹, (Member, IEEE), MOHAMED ADEL SERHANI², FARAG SALLABI¹, (Member, IEEE), KHALED SHUAIB¹, AND HEBBA M. KHATER^{1,3}

¹College of Information Technology, United Arab Emirates University, Al Ain, United Arab Emirates

²College of Computing and Informatics, University of Sharjah, Sharjah, United Arab Emirates

³Faculty of Computers and Informatics, Zagazig University, Zagazig 44519, Egypt

Corresponding author: Ezedin Barka (ebarka@uaeu.ac.ae)

ABSTRACT Organizational cybersecurity relies heavily on security operation centers (SOCs) to protect businesses and institutions from emerging cyber threats. In recent years, the complexity and sophistication of cyber threats have increased, pushing SOCs to their limits. As a result, SOCs struggle to address the evolving threat landscape due to their reliance on isolation technologies and reactive strategies. However, advanced technologies, such as artificial intelligence (AI) and machine learning (ML), have the potential to revolutionize SOCs by enhancing threat identification and response capabilities, as well as predicting and preempting risks. To address these challenges and highlight the full potential of SOC, this study provides a detailed overview through a comprehensive literature review that identifies gaps in existing research and examines the latest technologies used in the SOC environment to help address different operational and technical challenges and bring out their capabilities. Various methods, ranging from automated incident response and behavioral analytics to neural networks and deep learning, have been classified and compared. In addition, an in-depth reference architectural model, which is a blueprint for SOC integrating AI and ML into SOCs, is introduced. The proposed model provides a structured framework for implementation and offers insights into different SOC components and their interactions. Moreover, this systematic review emphasizes the benefits of these technologies for enhancing security operations. Finally, a case study is presented to describe the function of ML- and AI-powered SOC components to achieve optimum security. This paper concludes by discussing additional challenges and future research directions that may help advance the cybersecurity sector and provide insights into improving SOCs.

INDEX TERMS Artificial intelligence, cybersecurity, cyber threats, healthcare security, incident response, machine learning, neural networks, next-generation SOC, security operation center, threat detection.

I. INTRODUCTION

Cybersecurity has witnessed rapid advancements as threats across organizations in complex domains have increased. Consequently, Security Operation Centers (SOCs) are central to the defense architecture of an organization for monitoring, detecting, and responding to such threats. As current information technology systems pose complexities on traditional SOCs, they are being integrated with disruptive technologies

such as artificial intelligence (AI) and machine learning (ML) [1]. Recent AI and ML advancements have revolutionized the potential of SOCs by offering tools to augment capabilities by automating routine tasks and innovatively analyzing abundant security data. For instance, AI-based systems use complex algorithms to considerably enhance cyber threat detection and classification by searching for patterns that may be missed during manual analysis [2]. SOCs have various functions beyond threat identification. Modern-day SOCs are increasingly adopting proactive AI-based threat intelligence systems to analyze external and internal threat data.

The associate editor coordinating the review of this manuscript and approving it for publication was Tiago Cruz¹.

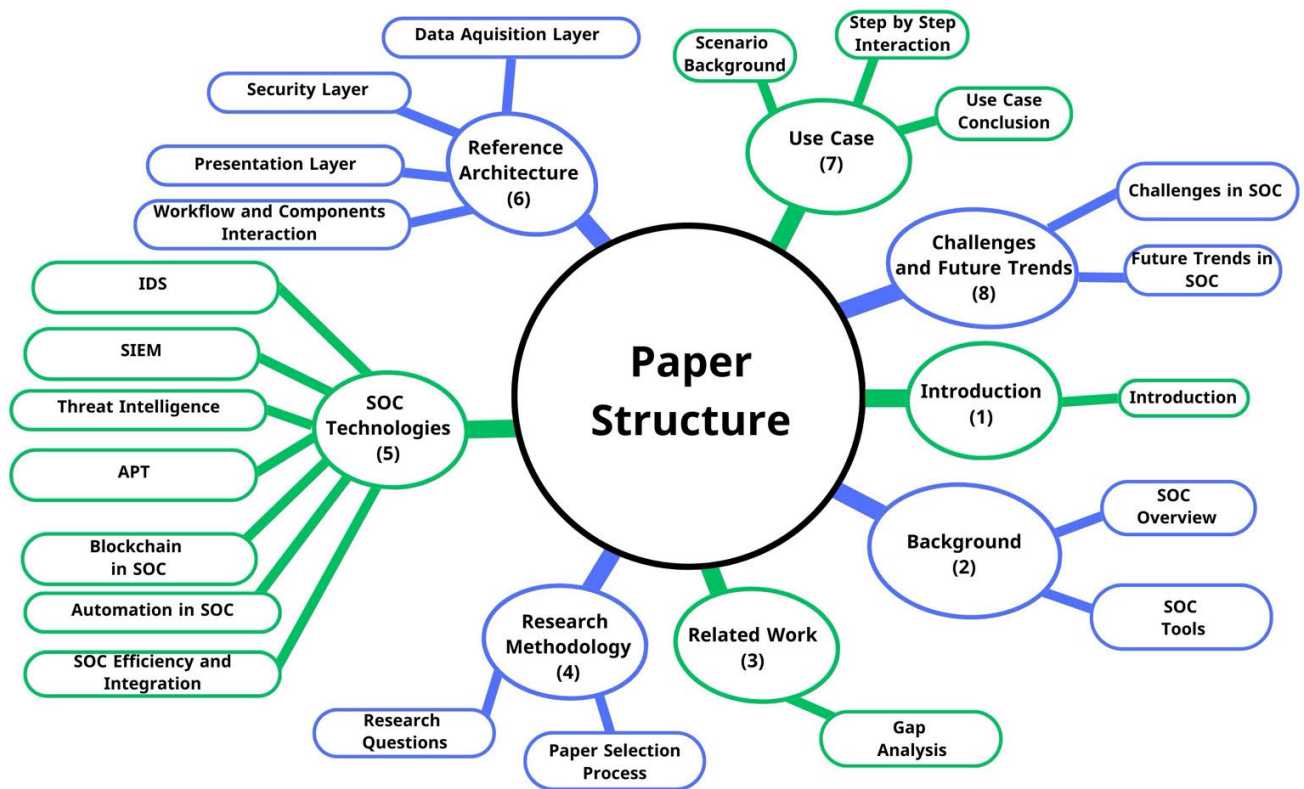


FIGURE 1. Overall paper structure.

These systems predict and foresee an attack. These systems are crucial and employ more sophisticated and continuous approaches to prevent advanced persistent threats (APTs) [3].

Despite these developments, adopting AI and ML into SOC operations has been problematic due to issues such as data overload, continuous updating of models, and complexity in managing AI-driven systems. Therefore, a comprehensive architectural model that can seamlessly merge these technologies within the existing SOC framework is required to enhance its effectiveness and ensure seamless functioning of components to prevent threats [4]. The reference architecture developed based on this review paper will have leveraged AI and ML to provide a framework wherein all SOC components, from intrusion detection and response to threat intelligence and APT detection, are fully integrated and optimized. It will be a benchmark that allows an organization to develop or enhance its SOC capabilities and empower them against complex and dynamic cyber threats.

This paper provides a detailed and current overview of SOC by effectively comparing different SOC components and realizing their functions. The following research problem was addressed: How can we design and implement a next-generation SOC that will effectively address the evolving challenges and threats against modern IT environments.

We contribute to this question by covering the following aspects:

1. **Comprehensive Overview:** The current technologies used at SOC are reviewed in depth, for which a transparent classification model that maps various tools and innovations related to SOC is required.
2. **Building a Reference Architecture:** Our aim is to develop a reference model, in which all SOC components work seamlessly and complement each other to ensure robust security operation. This model will highlight how different SOC elements, such as threat detection, incident response, and threat intelligence, can be integrated and enhanced using AI and ML, demonstrating such building blocks can collaborate to maximize the overall impact and efficiency of SOC.
3. **Healthcare Use Case:** We discussed the applicability of the proposed SOC model in the healthcare sector to protect sensitive data and enhance overall security. We also provide a practical example on the implementation of these ideas.
4. **Understanding Challenges and Opportunities:** The most salient challenges and opportunities in SOC were identified to identify areas wherein SOC can evolve and innovate to handle modern cybersecurity complexities.

Fig. 1 provides an overview of the structure of the paper. The remainder of this paper is organized as follows: Section II provides a background on SOC and their tools and security triage. In Section III, the literature on SOC is surveyed and systematically reviewed. Section IV discusses the methodological approach employed herein. Section IV-A provides a comprehensive literature review of technologies in SOC. Section V presents and discusses a reference architectural framework for SOC. Using a case analysis of healthcare intuitions, Section VI illustrates the application of Next Generation SOC. Section VII presents and examines the current and future issues as well as trends in SOC. The conclusion summarizes the findings of the Next Generation SOC and identifies their potential.

II. BACKGROUND

A. SOC OVERVIEW

SOCs are the core of any organization's cybersecurity. SOC is used for continuous monitoring, detection, and response against possible security threats. As cyber threats have become increasingly complex, SOC has undergone advancements in their roles and capabilities. A traditional SOC responsively identifies incidents and responded to them. However, with the onset of newfangled threats, SOC has had to become proactive in foreseeing and preventing such attacks via predictive analytics and threat intelligence as well as AI-driven automation [2]. At present, SOC combines various technologies and processes into a security posture that manages log analysis, threat detection, incident response, and compliance reporting. These changes were implemented as SOC has advanced from incident response and monitoring toward continuous adaptation to emerging threats.



FIGURE 2. SOC tools.

B. SOC TOOLS

The success of SOC depends on the tools employed for collecting, analyzing, and acting on security data. Herein,

the most prominent tools in SOC are discussed, which are crucial for the security strategy. Fig. 2 provides an overview of the different tools used in SOC.

1) SECURITY INFORMATION AND EVENT MANAGEMENT (SIEM)

SIEM systems provide a single aggregate point to combine and analyze security data derived from infrastructural components such as network devices, servers, and applications. SIEM tools correlate all aggregated events within the entire infrastructural organization, identify potential security incidents, and alert the analysts for further investigation [6].

2) INTRUSION DETECTION SYSTEMS (IDS)

IDSs are deployed to detect any unauthorized network access or attack. These systems are of two main types: signature-based, in which known threats are detected by matching patterns from a database of known signatures, and anomaly-based, in which deviations from normal behavior that indicate the existence of an unknown threat are detected [7].

3) THREAT INTELLIGENCE PLATFORMS

Threat intelligence platforms are integrated into SOC to enrich security alerts with context and improve their decision-making process in response to incidents. Threat intelligence platforms gather data from global threat databases, dark web forums, or internal logs; analyze these data to track emerging threats and vulnerabilities; and send out notifications [8].

4) SECURITY ORCHESTRATION, AUTOMATION, AND RESPONSE (SOAR)

SOAR platforms were recently added to SOC to automate security incident response and streamline workflows. Such tools integrate with other technologies in the SOC such as SIEM systems and IDSs to perform routine tasks such as alert triaging, threat containment, and remediation [9].

5) ADVANCED PERSISTENT THREAT (APT) DETECTORS

APT detectors are specialized tools developed to identify and mitigate long-term, targeted cyber-attacks. Commonly advanced APTs can bypass traditional security measures and often linger in the cyberinfrastructure without exposure. APT detectors combine behavioral analysis, anomaly detection, and threat intelligence to identify subtle cues of an APT [3]. They help the SOC take proactive steps to block the full materialization of severe APTs by heuristically detecting the first symptoms from network traffic and user behavior recordings.

C. SOC TRIAGE PROCESS

The SOC triage process is an integral part of incident management. It involves reviewing and prioritizing security alerts in a structured manner to ensure that the most meaningful threats are resolved first. The process typically progresses via

several key stages, each enabled by different SOC tools and technologies. Fig. 3 provides an overview about the Security Triage process.

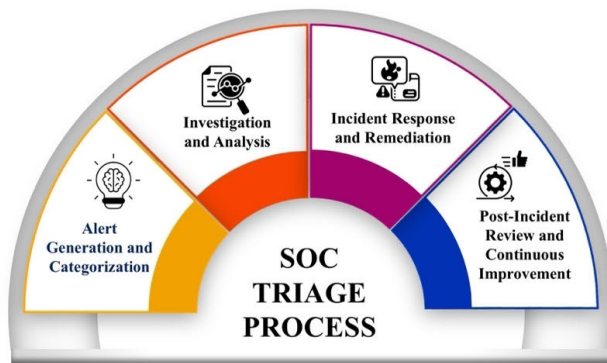


FIGURE 3. Security triage.

1) ALERT GENERATION AND CATEGORIZATION

The triage process initiates when alerts are generated from SOC tools such as SIEM systems, IDSs, and threat intelligence platforms. These different alerts are then classified based on severity, targeted assets, and possible organizational impact. AI and ML are employed to automatically filter out false positives and prioritize the alerts that need attention for immediate action [10].

2) INVESTIGATION AND ANALYSIS

After alerts are categorized, SOC analysts investigate the most severe incidents. This may entail data collection from other departments, including network logs, data from endpoints, and threat intelligence feeds, to further understand the nature of the threat. For instance, an alert created by an anomaly-based IDS may have to be analyzed in depth to ascertain whether or not it represents a new, hitherto unknown threat [11].

3) INCIDENT RESPONSE AND REMEDIATION

In the post-investigation phase, the SOC must determine an accurate remediation of the identified threat. This may involve isolating systems, patching, or executing other pre-defined incident response plans. Many such response actions can be automated using the SOAR platform for speedy and completely managed deployment to contain and remediate threats [12]. Finalizing the triage involves documenting the incident and response activities undertaken for compliance and future reference.

4) POST-INCIDENT REVIEW AND CONTINUOUS IMPROVEMENT

After resolving an incident, a post-incident review is held to assess the effectiveness of the SOC response. This includes root cause analysis of the incident to understand if the response could have been more effective as well as measures to improve the future SOC operations [13]. Furthermore,

SOC personnel should undergo continuous training, and these programs must be updated to keep them informed about new, more advanced threats [14].

III. RELATED WORK

Different aspects of SOC have been discussed in various systematic Literature Reviews (SLRs) published over the last few years. These studies have considerably contributed to addressing numerous problems related to SOC such as behavior, situation awareness, and complex cyber threat detection. However, most reviews have focused on specific parts of SOC, frequently ignoring the requirement of providing a general framework that includes all critical components. These studies are compared with this review in terms of their focus, methodologies, contributions, and limitations.

Goodall et al. [15] focused on explaining suspicious behaviors within SOC, providing some fundamental understanding of how SOA analysts can recognize and interpret suspicious activities. The methodology was systematic and used case studies to describe the types of behaviors that should alert an SOC. Goodall et al.'s [15] overly specified behavior identification. They provided valuable knowledge in the field, but their review did not contribute to general SOC operations, particularly the technological integration and comprehensive framework of a functioning SOC. Ofte and Katsikas's [16] SLR primarily focused on the situational awareness in SOC and emphasized that the security environment must be maintained intact to enable effective SOC operation. They determined how SOC analysts perceive security incidents and studied their consequent reactions to those perceptions to determine timely and correct decision-making. Although they made some valuable contributions, their review focused only on human factors and did not discuss technical or architectural challenges. As a result, the interaction of situational awareness with technologies and the overall SOC architecture were neglected.

Akinrolabu et al. [17] focused on detecting advanced cyber-attacks in SOC. They presented a survey on threat-detection methodologies, mainly focusing on how SOC can detect new and unknown threats. Their approach was timely, particularly concerning the present-day complexities and challenges in ensuring cybersecurity. However, their research was narrow, as they did not focus on SOC component integration. Although the review provided deep insights into threat detection, holistic solutions involving constructive interaction between different SOC technologies were not reported. Ben-nouri et al. [18] explored the role of SOC in ensuring the security of IoT systems, a rapidly growing area of concern in cybersecurity. They identified specific challenges with SOC in securing IoT environments such as managing a vast number of connected devices and unique device vulnerabilities. Although they provided valuable insights, it was a relatively niche study that did not generalize findings to broader SOC environments. This focus potentially reduced the generalizability of the study to traditional SOC operations. Spring and Illari [14] reviewed human decision-making processes on

computer security incident response in SOCs. They examined the actual decision-making processes of SOC analysts under pressure, detailing their cognitive reasoning. Human factor is essential in understanding the role of analysts within SOCs. However, they did not integrate the process with SOC technologies or propose a general architectural model of human and technological elements.

Sworna [19] analyzed natural language processing (NLP) techniques for host-based IDSs and demonstrated their integration in an SOC to improve security operations. Textual data analysis highlighted the potential of NLP in enhancing intrusion detection accuracy and efficiency. However, they focused only on NLP and did not consider its integration with other SOC technologies and processes to present a more comprehensive security solution. Neupane [20] focused on explainable AI (XAI) in IDSs and conducted extensive studies on AI-driven SOC operations to address the need for transparency with accountability in such systems used for security purposes. Their study contributed to the understanding of possibilities of AI interpretability for SOC analysts by reviewing various XAI techniques. However, the study is limited because the findings are not generalized and the conclusions obtained for other components of broader SOC architectures are not integrated.

Dun et al. [21] studied the current research trends concerning emerging technologies for state-of-the-art SOCs. They

briefly described how new tools and technologies could further enhance the capabilities of SOCs in dealing with advanced cyber intrusions. Although they made significant contributions toward the future of SOC technologies, they did not consider an integration framework for such emerging technologies within an SOC architecture. Although the survey supported future directions, it did not provide a complete road map for SOC implementation. AfzaliSeresht et al. [22] presented a general SOC framework in their systematic review to showcase basic SOC operations with a broader perspective on how SOCs should be organized. The framework was helpful but shallow from several perspectives such as integration of AI/ML technologies and effective interaction among different parts of an SOC. This gap further decreased the potential for practical application of the framework in modern SOCs where AI/ML integration is crucial.

A. GAP ANALYSIS

Table 1 summarizes some critical contributions and other vital features of previous SLRs on SOCs. For instance, Goodall et al. [15] dealt with suspicious behavior in SOCs but did not underpin it with a broader architectural framework. Ofte and Katsikas [16] dealt with situational awareness but limited their focus on human factors and did not cover more generic technical and architectural issues. Akinrolabu et al. [17] provided insights into detecting

TABLE 1. Overview of related surveys on SOC.

Study	Year	Focus	Methodology	Contributions	Limitations
[15]	2019	Identifying and explaining suspicious behavior in SOCs	Systematic review, case studies	Offers key insights into suspicious behavior in SOCs	Focuses only on behavior identification, lacks a broader SOC framework
[16]	2023	Enhancing situation awareness within SOCs	Systematic review, human factors analysis	Highlights the importance of situation awareness in SOCs	Limited to human factors, does not tackle technical or architectural challenges
[17]	2018	Detecting sophisticated cyber-attacks within SOCs	Systematic review, threat detection methodologies	Discusses the challenges in identifying advanced threats	It does not propose a comprehensive solution for integrating SOC components
[18]	2023	The role of SOCs in securing IoT devices	Systematic review, IoT-specific challenges	Identifies challenges specific to SOCs in the IoT context	Focused on IoT, lacks broader application to general SOC environments
[14]	2021	Human decision-making in security incidents	Review, decision-making models	Explores how decision-making processes unfold within SOCs	Focuses on human factors, does not address technological integration
[19]	2023	The use of NLP methods in host-based intrusion detection	Review NLP techniques	Discusses the application of NLP in intrusion detection systems	Focused on a specific application, lacks a broader SOC context
[20]	2022	Explainable AI in intrusion detection	Review, explainable AI	Reviews techniques for explainable AI in intrusion detection systems	Limited to explainable AI, lacks a broader architectural focus
[21]	2021	Next-gen SOC technologies	A Systematic Review, Emerging Technologies	Reviews emerging technologies for next-gen SOCs	It does not provide a comprehensive integration framework for SOC components.
[23]	2022	General SOC framework	Systematic review, framework analysis	Proposes a general SOC framework	Does not delve deeply into AI/ML integration or component constructive interaction
Our SLR	2024	Comprehensive SOC framework with AI/ML integration	Systematic review, reference architecture, use case analysis	Proposes a detailed reference architecture, identifies challenges, provides a use case	Focuses on component constructive interaction and AI/ML integration and addresses operational challenges

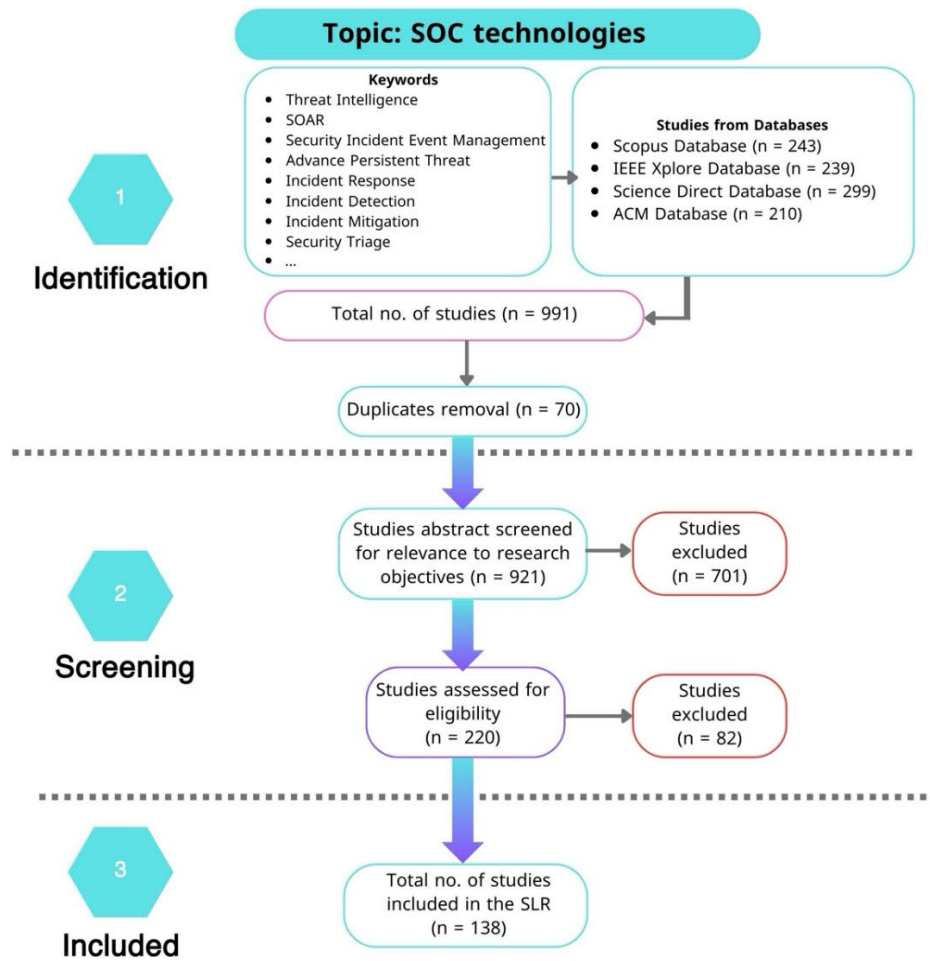


FIGURE 4. PRISMA-based SLR methodology flow diagram.

sophisticated attacks but did not propose a holistic solution that integrated various components of an SOC. Bennouri et al. [18] focused on SOC challenges within IoT and did not generalize these findings to SOC within other contexts. Sworna [19] elucidated human decision-making processes and applied NLP for intrusion detection. However, neither study provided a general view of SOC operations. Neupane [20] discussed XAI in IDSs but did not have a broader architectural focus. Dun et al. [21] surveyed emerging technologies for next-generation SOC, and AfzaliSeresht et al. [22] made a generic proposal for SOC. However, integration with AI/ML and constructive interaction between the components of SOC were not studied in detail in these reviews.

Cyber threats have become more complex and sophisticated, pushing traditional SOC to their maximum limits. This is evident from the scenario that current SOC cannot deal with an evolving threat landscape because they depend on isolation technologies and reactive strategies. Moreover, advanced technologies such as AI and ML can revolutionize SOC by identifying and responding to threats more effectively. These technologies can also predict and preempt such risks. Although AI and ML are promising, existing stud-

ies have not provided a holistic and integrated approach to exploit AI and ML in all SOC components.

IV. RESEARCH METHODOLOGY

Systematic research review approaches based on the literature on technologies, classifications, challenges, and solutions of SOC were employed herein. Unlike an unstructured review, a systematic literature review guarantees a comprehensive and unbiased analysis. The SLR approach is based on a predefined protocol of reviews that conforms to standards of the PRISMA framework [24]. We aim to conduct a study with the refinement and scientific rigor required for collection, evaluation, documentation, and synthesis of findings. Subsequently, we comprehensively and reliably studied the available literature on SOC to assess their capacity and potential developments suggested therein.

A. RESEARCH QUESTIONS

Four research questions were created to guide the investigation and provide an evidence-based overview of SOC. They formed the basis of this study and research methodology,

TABLE 2. Inclusion and exclusion criteria.

Inclusion Criteria	Exclusion Criteria
- Peer-reviewed journal articles, conference papers - Published between 2018 – 2024 - Written in English - Related to the field of SOC - Presenting original research, case study, or empirical studies	- Non-peer-reviewed sources - Published before 2018 - non-English language articles - Irrelevant topics to SOC - Non-peer-reviewed articles, opinion pieces, editorials, and non-scholarly publications

thereby guiding the selection of relevant literature for a comprehensive analysis.

- **RQ1:** What are some of the critical technologies in use within an SOC that can enhance security operations?
- **RQ2:** What architectural model is the most valuable in safeguarding organizations, particularly that driven by AI and ML?
- **RQ3:** How can the proposed next-generation SOC be integrated constructively within the healthcare system to safeguard sensitive information?
- **RQ4:** What are some of the most paramount challenges or continuing concerns faced by SOC today?

B. PAPER SELECTION PROCESS

The selection process for the systematic literature review comprised two stages:

Initial selection: We conducted an automated search with keywords, titles, and abstracts in major academic search engines such as ACM, ScienceDirect, Scopus, and IEEE Xplore. This initial search included conference papers, journal articles, and book chapters. After removing duplicates, we obtained **991** unique articles.

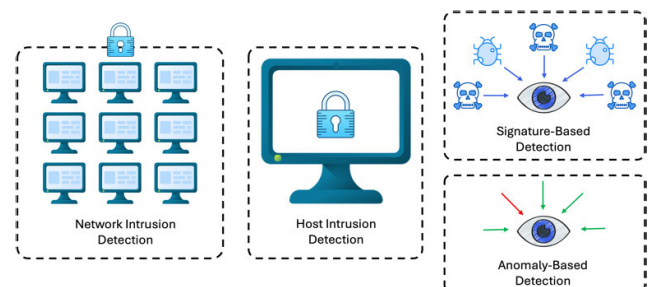
Final selection: For more precise selection, we tested the articles against the criteria listed in Table 2. Then, we read the full text of selected papers, carefully testing their quality and relevance. Only those papers clearly describing their evaluation procedure and providing details were considered for the final selection. After such evaluations, **138** appropriate papers were taken for further qualitative analysis. Fig. 4 shows the PRISMA flow diagram of the SLR methodology employed herein.

V. SOC TECHNOLOGIES

In this section, we address **RQ1** by analyzing various technologies, including IDS, SIEM, Threat Intelligence, APT, Blockchain in SOC, automation, and SOC efficiency and integration.

The complexity of cybersecurity makes the public incapable of understanding most of its elements. The SOC may be broadly defined as a well-trained and devoted team whose goals are to improve an organization's security on a constant

basis. It is basically responsible for preventing, detecting, analyzing, and responding to cybersecurity incidents through advanced technology and processes and procedures in place. Essentially, an SOC works like a command center or like the nerve center that oversees the protection of an organization from cyber threats to ensure business continuity and operational resilience [18]. Fig. 5 shows the basic anatomy of a SOC.

**FIGURE 5.** SOC technologies.**FIGURE 6.** Intrusion detection system types.

A. IDS

IDSs are crucial for ensuring cybersecurity, particularly in large, complex network environments. Related studies have evolved to address the increasing complexity and volume of cyber threats and contributed by proposing novel methodologies, tools, and frameworks. Fig. 6 shows the different IDS types. Yang et al. [25] explored Bayesian deep learning models for improving uncertainty quantification in anomaly detection. The proposed approach enhanced the trustworthiness of IDS alerts by jointly modeling aleatoric and epistemic uncertainties, which reduced false positives, a common issue in traditional IDS. This improvement was validated using UNSW-NB15 and CIC-IDS-2017 datasets, demonstrating the efficacy of the model in providing more reliable alert predictions. Although the approach showed promise, the

complexity of the Bayesian models and computational overhead posed potential limitations.

Complementing this, Kwon et al. [26] introduced VISNU, a visualization system optimized for centralized security operation centers (CSOCs). VISNU helped security experts manage the overwhelming volume of alerts by classifying and visualizing them based on patterns and history, thus enabling a more effective analysis of abnormal activities. This method focused on real-time monitoring although it primarily addressed visualization rather than the underlying detection accuracy, thereby limiting its effectiveness when dealing with novel or sophisticated attacks. Song et al. [27] further explored visualization techniques using VisIDAC, providing a 3D representation of security events across multiple networks. This tool aided in understanding the flow and trends of security events, making it easier for operators to identify large-scale attacks such as distributed denial-of-service (DDoS) attacks. Although 3D visualization improved the comprehension of such events, they could overwhelm inexperienced operators in such environments and limit its usability across different expertise levels.

Wang et al. [28] proposed a transformer-based framework for alert aggregation and attack prediction to address the issue of excessive and redundant alerts as well as improve alert handling. By aggregating similar alerts and using advanced ML models, the framework enhanced prediction accuracy and reduced the cognitive load on security analysts. However, the reliance on high-quality training data and computational demands of transformer models can present challenges in real-world applications. Sen et al. [29] focused on detecting multistage cyberattacks in intelligent grids by leveraging contextual correlation. Their approach automated the process of correlating cyber intelligence, thus reducing manual effort and enhancing the effectiveness of security operations. Despite these advancements, accurately assessing and understanding cybersecurity threats in dynamic environments remain challenging. Vaarandi and Guerra-Manzanares [30] explored stream clustering-guided supervised learning for classifying network IDS (NIDS) alerts. Their ML framework effectively reduced false positives, easing the workload of security analysts. However, the dependency of the framework on high-quality labeled datasets for training remains a significant limitation, highlighting the need for more robust data collection methods.

Chavali et al. [32] introduced two reinforcement learning (RL) methods, namely TD3-AP and SAC-AP, for alert prioritization in IDS. These methods addressed the instability and limited exploration capabilities of previous models, resulting in more efficient alert management. Although these approaches demonstrate significant improvements, their reliance on computationally intensive models may pose challenges for organizations with limited resources. Ring et al. [33] focused on host-based IDSs (HIDS) and proposed the ALAD algorithm, which used deep learning to analyze sequence-to-sequence behavior in system calls.

The algorithm outperformed traditional methods, but the “black box” nature of deep learning models limited their interpretability, posing challenges for analysts who require transparency in decision-making processes. Ngo et al. [34] proposed a heterogeneous hardware-based framework for NIDS, leveraging lightweight neural network models to enhance detection accuracy and minimize power consumption in the context of IoT security. Although their framework achieved impressive results, the high computational cost of deploying such systems on resource-constrained IoT devices remained a significant limitation.

Vaarandi and Guerra-Manzanares [36] applied active learning techniques to NIDS alert classification and addressed a research gap by involving human experts in the ML model lifecycle, thereby improving the classification of high-importance alerts. Despite their potential benefits, relying on expert input may limit the scalability of their approach in large organizations. The growing interest in XAI in IDS was exemplified by Neupane [20] who surveyed current methods, challenges, and opportunities in X-IDS. They highlighted the need for models that clearly explained their predictions, crucial for building trust among SOC analysts. However, balancing explainability and model performance remains a significant challenge. Mézešová et al. [37] proposed a methodology for evaluating attackers’ skill levels in multistage attacks, which provided better defense strategies. Their approach relied on correlation rules to reduce false positives and improve alert accuracy. Although effective, the reliance of their method on predefined rules may limit its adaptability to new and evolving attack vectors. Several studies have focused on enhancing decision support tools for SOC analysts. Happa et al. [38] developed a visualization tool that helped analysts prioritize response strategies by relating network alerts to business processes. Similarly, Schraagen et al. [39] emphasized on balancing broader operational context with detailed investigation during incident management. Both studies highlighted the need for tools that integrated technical and business considerations, although their practical implementation in diverse organizational contexts may present challenges.

Wawrowski et al. [40] explored the application of XAI-based explainability in network traffic monitoring for understanding and interpreting the results of anomaly detection methods. Their work underscored the importance of transparency and interpretability in IDS, particularly in enhancing the trust of security analysts in automated detection systems. However, the complexity of integrating XAI techniques into existing systems and workflows remains a significant issue. Engel et al. [41] addressed the issue of customizable operation centers, presenting a dashboard system where users can personalize their security monitoring experience, and SOC analysts can enhance their efficiency by focusing only on information related to them. However, the effectiveness of the system in a large-scale and dynamic environment with many alerts must be assessed.

Kanemoto et al. [42] proposed a scheme to detect successful attacks by remotely emulating shellcodes based on IDS alerts. Their approach enhanced the precision and speed of attack detection but its practical applicability in resource-restrictive environments is limited due to resource-intensive emulation. Considering deep transfer learning in IDS, Latif et al. [43] proposed an optimized framework in IIoT networks, which exhibited excellent detection performance; however, its reliance on computationally expensive models affects its feasibility in resource-constrained scenarios.

Jaroensiripong et al. [44] used an image classification model within an IDS framework to convert network data flows into images using the Hilbert curve mapping algorithm. Their approach is promising in improving detection accuracy, but the added computation overhead due to image processing could be a limiting factor for its use in real-time applications. Presekal et al. [45] conducted forensic analysis on graph deep learning-based OT networks. Although the location of a cyberattack could be determined using their approach, graph-based models are assumed to be complex. Therefore, they are not scalable and cannot be applied in real-time. Wu and Moon [46] suggested an alert correlation approach for cyber-manufacturing systems comprising cyber and physical alerts. Their approach considerably decreased false positives and increased the general efficiency of IDS in such environments; however, dealing with heterogeneous alert types in a less structured environment was considerably complex. This problem of alert prioritization for cyber SOC was addressed by Shah et al. [47] who introduced a two-stage process for optimizing alert selection. Although their approach improved the time within which an alert can be considered for investigation, the dependency on organization-specific parameters might limit the generalizability of their approach.

Zhong et al. [48] addressed a critical need for effective collaboration between cybersecurity analysts, for which they proposed the AOH-Map for mind mapping that supported the collaborative analysis of cyber securities. This system made information sharing more convenient and facilitated better task allocation among analysts during threat detection and response. However, it has scalability issues because it depends on large SOC that require manual intervention. Javeed et al. [49] suggested an intelligent deep learning-based IDS for software-defined networking architecture in smart consumer electronic networks. When submitted by users, it is highly accurate; this method was effective in detecting multiple types of attacks. However, its integration into existing infrastructures remains complicated. Vermeer et al. [50] researched SOC workflows, particularly the management of NIDS rules, and highlighted some crucial factors that impacted SOC workflows: rule specificity and quality of alerts. They also presented some recommendations for optimizing SOC workflows to efficiently detect network attack intrusion. Their study was based entirely on interviews with professionals and therefore their outcomes

could not be generalized. Vo et al. [7] proposed an AI-based IDS approach that unified signature-based detection and deep analysis. Although this approach enhanced the speed and quality of detection, deep learning models posed high computational demands. Jiang and Atif [51] used a cognitive cybersecurity enhancement approach toward capabilities in vulnerability assessment by integrating human expertise with ML. Although this approach is promising, data-intensive preprocessing may pose a limitation.

Pérez et al. [52] proposed a new IDS methodology that combined multiplex networks and time series analysis. Such an approach limited the number of generated alerts generated as it focused on specific IP addresses and predefined time windows, but its effectiveness in a dynamic context needs validation. Kalakoti et al. [53] proposed an active learning model for IoT botnet identification using explainability techniques such as LIME and SHAP. Although this model added posed detection capability, its dependence on labeling data and the complexity integrating explainability method can limit its real-time applicability. Hamad et al. [55] proposed REACT, an autonomous intrusion response system for intelligent vehicles. Their system improved a vehicle's response to cyber incidents, but the embedded platforms may limit its adaptability to other vehicle models. Shah et al. [56] built adversarial attacks based on RL, which were used to test and iteratively improve the strategies of the defender. Although this methodology provided insights into the operationalization of IDS for resilience to adversarial attacks, the complexity of the adversarial models may hinder their practical applicability.

Mocanu and Scripcariu [57] discussed the case study on implementing an SOC using open-source tools such as Security Onion. Although their approach was economically viable for organizations, the challenges associated with resource utilization and skill gaps must be addressed. Shon et al. [58] discussed a semisupervised alert filtering method, which minimized alert fatigue by clustering similar alerts from names and IP addresses. Their approach enables generating highly accurate filtering alerts; however, its reliance on predefined rules might limit its flexibility. Goodall et al. [15] described Situ, a visual analytics application aimed at helping analysts identify abnormal activities in network datasets. Their application enhanced situational awareness by providing additional context about detected anomalies; however, its integration with existing processes can be challenging. Agyepong et al. [59] outlined a framework for measuring the performance of SOC analysts based on key performance indicators such as analysis quality and accuracy of reports. Although their framework could comprehensively measure performance, the lack of a consolidated agreement on the core roles of analysts remains an issue. Sworna [19] provided a systematic review on the role of NLP for HIDS enhancement, highlighting the increased adaptation of newly proposed NLP methodologies for vulnerability detection on a large dataset. NLP was employed for detecting intricate attacking patterns, thereby refining the performance

of HIDS on SIEM frameworks with low false alerting and high detection accuracy. Eriksson and Grov [60] conducted a human-centered study on the usage of XAI in SOC, concentrating on SHAP and LIME. They reported that XAI could improve analysts' trust in their machine-learning models, but its complexity may be a barrier. Prabu and Sudhakar [12] proposed an automated intrusion detection and prevention model that improved threat assessment and security data solutions. Although their model yielded highly accurate results, the problems in assimilating such systems into the existing infrastructure may impact their performance. Hiruta et al. [61] proposed a method for prioritizing IDS alerts based on traffic behavior, considerably reducing the number of high-priority alerts to be investigated by SOC analysts. Their process is effective; however, relying on traffic behavior analysis may limit its adaptability to rapidly changing network environments. Table 3 provides an overview for the IDS related paper in SOC.

The analyzed studies reveal that IDS today face significant challenges in effectively managing the increasing complexity and volume of cyber threats. Advancements such as Bayesian deep learning, reinforcement learning, and visualization tools have shown promise in enhancing the detection accuracy and alert management, as well as issues related to computational overhead and reliance on high-quality training data. Moreover, the effectiveness of these systems is often compromised by their interpretability. Many advanced models operate as black boxes, making it difficult for analysts to trust their predictions. Additionally, the integration of explainable artificial intelligence (XAI) techniques is critical for building this trust. However, the complexity of these methods can limit their practical application within SOC. Although these innovations are vital for advancing cybersecurity defenses, their successful implementation requires addressing the limitations of scalability, adaptability, and the need for transparency in decision-making processes.

B. SIEM

Changing dynamics in cybersecurity have placed SIEM systems at the heart of organizational defense mechanisms, particularly in SOC. These systems collect, analyze, and respond to security events. Fig. 7 shows the SIEM process flow. The literature on research and development in SIEM technology indicates that the advancement of these systems has become an integral part of the ever-increasing challenges posed by modern cybersecurity threats. Complementary to such advancements, Kim and Kwon [62] optimized the performance of deep learning models in threat classification for SIEM systems. They explored models such as CNN, LSTM, and GRU; the latter became more precise in threat classification. However, they also mentioned that CNN-static (1D) ensured the most cost-effective balance of accuracy and computational efficiency, particularly in real-world SOC environments. This performance and efficiency balance becomes highly critical as an SIEM system needs to process

large amounts of data. Similarly, Bikov et al. [63] mentioned that intelligent SIEM solutions should be combined with threat-hunting techniques for enhancing the efficiency of SOC and reducing the security costs, making these systems much more efficient in the detection and mitigation of threats.

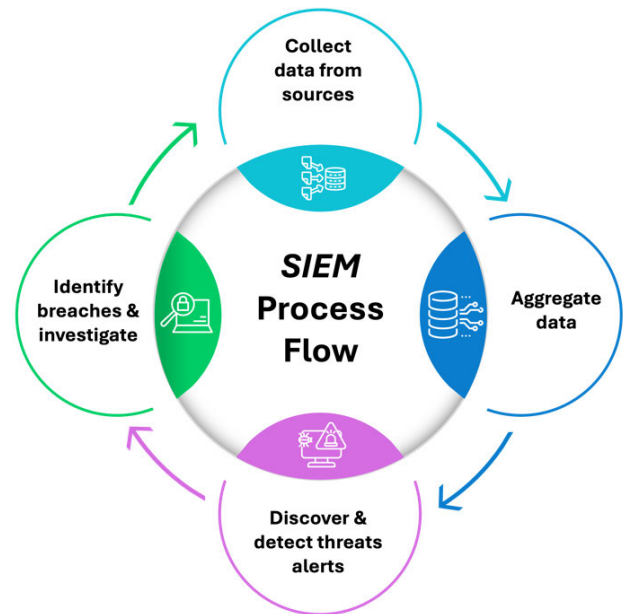


FIGURE 7. SIEM functions overview.

Axon et al. [64] proposed sonification within SIEM tools for network security monitoring and demonstrated that it enhanced monitoring performance for some tasks, thus indicating a potential inherent value of alternative data representations to improve situational awareness. Using this approach, data are converted into sound and enhanced the monitoring tasks of security practitioners by strengthening their capabilities to detect anomalies and respond to security incidents. Sonification in SIEM systems can also be employed to improve situational awareness of an SOC. However, the operational usage of sonification in SOC needs further validation. Akinrolabu et al. [17] emphasized that sophisticated cyber-attacks surpassed classic network security measures and showed that modern SIEM systems could not cope with detecting multistage attacks as most features were connected with concrete malware structures. These issues can be overcome by enhancing SIEM systems with inputs from experienced SOC analysts and embedding more effective detection mechanisms within them.

SIEM systems are widely used in IoT and smart city technology. Alharbi [35] addressed alert fatigue by developing an extended isolation forest model that identified and highlighted anomalous alerts. Their model reduced the number of alerts by over 80%, considerably easing the burden on SOC analysts. However, the effectiveness of this approach in highly dynamic environments, where alert patterns may constantly change, requires further investigation. Arfaoui et al. [65] first proposed SOC-as-a-Service (SOCaaS)

for IoT applications by developing an architecture for IoT network risk detection and mitigation. However, such systems experienced difficulties in scalability when implemented in constrained-resource environments. Shah et al. [54] addressed the fair distribution of CSOC resources by developing a dynamic weighted alert queuing system. This approach ensured fairness in the alert examination; however, its complexity may pose issues when implemented on different organizational levels. Bennouri et al. [18] highlighted the importance of SIEM systems in securing IoT devices and reviewed contemporary challenges in the security of IoT devices and their critical dependencies on SOC capabilities to mitigate threats. However, resource constraints in IoT devices may hinder the implementation of traditional SIEM architectures, necessitating the development of more lightweight and scalable solutions. Hwoij et al. [66] extended this idea to intelligent cities and proposed an SIEM architecture for monitoring and protecting large IoT devices in an innovative city environment, thereby emphasizing on round-the-clock monitoring and near-real-time response to security incidents. However, enhanced network bandwidth and data processing capabilities are required for deploying such systems at large scales. Mokalled et al. [67] examined the implementation of SIEM solutions in real-world situations and discussed the importance of choosing an accurate SIEM system for technical and organizational requirements. Their study offered a framework for assessing the applicability of SIEM systems such that organizations could choose appropriate solutions based on their unique requirements and security objectives.

Najafi et al. [68] presented MalRank, a graph-based inferencing algorithm enriching SIEM-based knowledge graphs for threat detection; it scored entities' maliciousness regarding their relations. The model achieved high detection rates, thus representing the value in SIEM systems for identifying previously unknown malicious entities. Sancho et al. [69] proposed security threat classification and risk estimation schemes using security event management data based on the bug bar approach in configuring threat priorities. Their model obtained enormous correlations using commercial SIEM systems. In other words, the incorporation of alternative models of threat classification may be substantial to existing SIEM systems but should be implemented with caution in operational environments. Bryant and Saiedian [70] suggested a kill-chain-based classification model for SIEM systems for improving their detection rates and removing false positives. They studied the implementation of a new log ontology and SIEM correlation rules, which showed positive results in the accuracy and descriptiveness of alerts. However, configuring this system accurately relies on complex and resource-intensive knowledge.

Chi et al. [6] addressed the issue of email customization missing from SIEM systems, particularly for SOC teams, to send customized reports to clients regarding their security. The proposed system outperformed existing SIEM software that have issues with email reporting because it was equipped

with options for email customization. The improved features in email templates for client reports enhanced the effectiveness of communication and efficiency of SOC activities. Skendzic et al. [71] explained the significance of monitoring and managing security events within an organization and showed how SIEM effectively monitored security incidents, particularly under modern remote working conditions, which posed another challenge to organizational security. McRee [72] investigated human factors influencing SIEM effectiveness by comparing visualized alert outputs with traditional text-based alerts. They found that security analysts preferred visualized alerts as it enabled faster and more accurate threat detection. Thus, SIEM systems must have user-friendly interfaces as information representation considerably impacts the ability of an analyst to respond to the threats at hand.

Hristov et al. [73] investigated the integration of Splunk Enterprise SIEM to identify DDoS attacks in IoT environments. Their solution epitomized the importance of SIEM systems in collecting and analyzing log data from diverse sources to improve real-time detection and response capabilities of the system in case of threats. SIEM systems are also being integrated with emergent technologies such as digital twins and big data platforms. Sankar and Fasila [75] integrated the Wazuh suite with the ELK stack to enable improved SOC functions. They transformed the security environment into an environment that could dynamically react via incident detection and response to improve the overall threat-detecting and incident response capability of SIEM systems. In contrast, researchers such as González-Granadillo et al. [8] focused on enhancing threat intelligence by integration with SIEM systems. They suggested that ETIP, which boosts the correlation and analysis capabilities of SIEM systems, integrated structured and unstructured threat data for enhancing the accuracy of detecting threats. However, ETIP relied on data from external sources, which may raise concerns over quality data and information overload.

Andrade and Torres [77] studied the extensive data analytics application of an SIEM system, primarily aimed at a computer security incident response team. They demonstrated the feasibility of the ELK Stack as a big data platform for security event detection, which allows big data tools to extend the functionalities of traditional SIEM systems. They also highlighted that scalability, and real-time processing may become bottlenecks in such systems, particularly in large enterprise environments. Cinque et al. [78] focused on the contextual filtering and prioritization of computer application logs for security situational awareness. Their approach could handle the problems arising from analyzing unstructured logs and considerably enhanced the efficiency of SIEM systems while monitoring critical industrial systems. Saraiva and Mateus-Coelho, [23] provided a systematic review of CyberSOC frameworks and found the requirement for a systematic framework that appropriately differentiated between actual and false threats. They shared insights into

the development of more practical SIEM systems to better support SOC analysts. Robert and Dimitoglou [79] developed a virtualized SOC by implementing a low-cost SOC infrastructure using virtualization technologies and open-source tools. They provided a scalable and efficient real-time network monitoring and threat-detection solution that benefited organizations with resource constraints. An ontology, CoCoA, was introduced by Onwubiko [80] to enhance cybersecurity operations within SOCs. Cybersecurity incidents can be managed in a structured manner using this ontology, which increases the analytical capability of SIEM systems and subsequently enables better threat detection and response. Khan et al. [81] studied the application of AI in threat hunting by proposing a cognitive and predictive tool to enable threat hunting using a minimum number of manual configurations. This AI-based tool identified unique network communication behavior to enhance proactive SOCs. However, such high reliance on AI and ML poses challenges, such as considerably large datasets, algo-bias, and poor threat-detection accuracy. Nursidiq et al. [82] proposed an enhanced threat-hunting framework to detect unknown threats to enterprise users. Their approach leveraged SIEM systems to create valuable intelligence, allowing them to devise new use cases and further fine-tune the security posture of organizations.

Ndichu et al. [83] discussed alert fatigue in SOCs and proposed an AI-assisted alert screening scheme that coupled ML with visualization tools for fast differentiation between actual threats and false alarms. With its high recall rate and low false-positive percentage, their method demonstrated good potential for supporting real-world applications and replacing partial human efforts for security analysts. BDR also formed the basis for a typical security operations dashboard, as described by Han [84]. This design will help SOCs to make operationalization easier via the abstraction of security activities and will therefore be seamlessly compatible with existing SIEM systems for added functionality. Collaboration within and across organizations has also been advocated to increase the effectiveness of SIEM systems. Mihindu and Khosrow-shahi [85] introduced the concept of a virtualized cybersecurity operation center that embeds collaborative visualization capabilities, enabling real-time security analysis of any given topic. Their study demonstrated the potential of such systems in increasing situational awareness and incident response. However, the cost and complexity of implementing such a virtualized environment may not be feasible for smaller organizations. Although practical, these models are highly data-dependent and require appropriate hyperparameter tuning; therefore, they often present some difficulties within dynamic environments. Islam [86] discussed the application of AI and ML to SOCs for further enhancing the detection and response capabilities of SIEM systems. However, they reported that such solutions were dependent on elaborate ML models that required large datasets and sophisticated infrastructure, hindering their wide applicability across organizations. Their research focused on

the continuous innovation of SIEM technology to adapt to the ever-emerging threat landscape.

Mohsen et al. [87] focused on aggregating target logs in SIEM systems to improve the detection of threats and reduce response time. They developed a more resilient SIEM framework that could efficiently handle increasing amounts of security data. Miloslavskaya [88] provided a historical perspective of SIEM systems, particularly for maintaining organizational compliance and security. Their analysis underlines the necessity of integrated SIEM solutions that can support security operations and intelligence collection. For instance, Miloslavskaya [88] identified the need for more integrated and effective SIEM systems to cope with the increased volume of data and threats to which organizations are exposed. As argued by Tounsi and Rais [89], threat intelligence must be presented in a standard format, and the reluctance of the organization to share information is a limitation to SIEM performance. More elaborative frameworks regarding sharing threat intelligence and collaboration must be developed to overcome SIEM limitations.

Calvo et al. [90] proposed a data science-based solution for the risk assessment of exposure in enterprise security, which depicted users and entities in the initial stages of the kill chain. Their framework improved the ability of SIEM systems to accurately predict threats, thereby improving their effectiveness in preventing such threats. Najafi et al. [91] laid the basis for the next generation of more intelligent and sophisticated SIEM systems and found that traditional SIEM systems have their fundamental roots based mainly on regulatory compliance. Such systems alone cannot help to detect advanced threat environments. The enhancements in the proposed architecture are expected to equip the current SIEM systems with state-of-the-art data science and engineering features to facilitate advanced threat hunting and pattern identification. They also highlighted the need for constant improvements in SIEM systems to fit the needs of rising threats. Wurzenberger et al. [92] similarly proposed a new technique for detecting anomalies in SIEM systems using the modification part of logs. Their variable type of detector identified the data types of log variables that helped detect anomalies that may be unrecognized using traditional approaches. Their method enhanced SIEM systems and enhanced their ability for intrusion detection in security events and matters. However, detailed log analysis is required to enhance the efficiency of SIEM systems. Table 3 provides an overview about the SIEM papers in SOC.

SIEM systems act as central components of organizational defense strategies, particularly within SOCs. These systems are designed to collect and analyze security events to enhance threat detection and response capabilities. Although advancements in deep learning and threat-hunting techniques have improved SIEM efficiency and reduced alert fatigue, challenges such as data overload and the need for real-time processing remain prevalent. Innovative approaches such as sonification have shown potential in enriching situational

awareness; however, their practical implementation requires further validation. Additionally, as organizations increasingly adopt IoT and smart city technologies, traditional SIEM architectures face scalability issues, necessitating the development of lightweight solutions. The integration of AI and ML has also been explored to enhance threat detection, although these methods often rely on large datasets and complex models, which limits their applicability. Furthermore, the importance of user-friendly interfaces and effective collaboration among analysts is highlighted as these factors significantly impact the performance of SIEM systems in operational environments.

C. THREAT INTELLIGENCE

Evolving cybersecurity demands have necessitated the integration of advanced technologies such as AI and ML to enhance threat intelligence and response capabilities. AI and ML have been introduced in cyber threat intelligence (CTI) to improve threat detection and response capability. Daniel et al. [133] presented a deep neural network model for tracing paths of investigation for cybersecurity incidents within SOC. They quantitatively assessed and measured the performance of security analysts in cyber investigations using the cyber analysis transition framework. Although the framework provided several important insights, its reliance on deep learning models highlighted the computational overhead and complexity associated with this approach. Demertzis et al. [93] presented a λ -architecture network flow forensic framework for social cognitive computing by developing ML rules for most SOC. This framework considerably improved their security and defense mechanisms against adversarial attacks. This framework used complex neural network models to analyze batch and streaming data, showing a paradigm shift from previous strategies for reactive security to proactive and automated defense systems. Although such frameworks have potential applicability, their development and practical adaptability to newly arising and evolving threats remain challenging.

Crooks and Vålán [94] focused on an easy-to-tailor CTI by emphasizing on AI and ML and proposed organization-specific threat intelligence system (OSTIS) centered around alert fatigue; this fatigue resulted from solutions that relied on data volumes rather than solutions tuned to the threat environment of a particular organization. However, this system can be inclined in ways that fail to consider far more significant threats that may not apply directly to some specific organization but might be highly dangerous. Based on these technological improvements, Forsberg, Frantti, et al. [134] identified a massive gap in the existing metrics used by SOC, which currently focused more on operational issues and less on technical performance. Therefore, they devised a new structure for developing technical performance metrics and performing evaluations related to the SOC's contribution to the overall cyber defense capabilities. Thus, their study is instrumental in developing a comprehensive set of metrics that can capture SOC operations.

Simran et al. [135] introduced AI shield and red AI frameworks that used ML to solve problem of CTI. AI shield operated based on real-time monitoring and threat intelligence fixation, whereas red AI employed adversarial simulations to enhance the defenses of AI systems. However, the use of AI in cybersecurity has several challenges such as AI failures and constant updating. Moreover, the frameworks proposed by Simran et al. can improve an organization's defense attitude toward advanced cyber threats. Pingle et al. [136] proposed a system based on deep learning to build cybersecurity knowledge graphs and support security analysts in detecting cyber threats. Using semantic triples to structure threat information for better decision-making in cyber investigations, the system verified the proof of concept of AI for enhancing SOC operations.

Irshad and Abdul [96] positioned rationality constraints within cyber defense, focusing on incident handling and threat intelligence. They reported that such updates to epistemic states are grounded on a partial understanding of the behavior of attackers but have inherent limits, which highlight the need of a more complete and systematic approach to threat intelligence that can provide fuller context to understand cyber threats. Kucsán et al. and Arikkat et al. [97], [137] applied NLP and large language models (LLMs) to automate the extraction of steps for direct data recovery from CTI reports. They showed that LLMs have a performance advantage over classical semantic approaches but produce high factual error rates; therefore, they must be refined before fully trusting their application in operational environments. However, balancing automation and accuracy will continue to be a challenge for AI-based threat intelligence.

Skias et al. [98] highlighted the need for a pan-European cybersecurity incident information-sharing platform (I2SP) to support cooperative detection and mitigation of cyber threats. This platform corresponded with the network information security directive of the European Union, and its cross-border collaboration provided a positive attribute for enhancing cybersecurity. Its success lies in organizations' willingness to share sensitive information by building trust among participants. Zibak et al. [131] proposed a success model to assess the effectiveness of CTIMP. According to the DeLone and McLean information system success model, identifying system quality, quality of information, and service quality as some of the critical success factors of these platforms. An organization can helpfully measure its investment in CTIMP using this approach. However, they revealed that this approach required constant updating due to the rapidly changing landscape of threats. Focusing on the human-centric dimension of cybersecurity, Ahmad et al. [95] presented a data mining model that transcribed security logs into natural language reports, thereby making the ensuing threats more understandable and interpretable for any SOC personnel. This is a more enlightening storytelling approach in cognition processes involved while processing security information and serves as an essential need toward making challenging data more accessible to humans.

Saddi et al. [139] discussed the role of generative AI for enhancing CTI. They indicated the use of generative AI in providing organizations with another layer of defense after vulnerabilities and potential threats, which otherwise could not be realized. Their findings further accentuate how AI will change the face of cybersecurity, with generative AI acting as an additional layer in SOC operation by quickly identifying threats and vulnerabilities. In contrast, it challenged explainability and trustworthiness in integrating generative AI within cybersecurity frameworks. Rastogi et al. [100] built a corpus by introducing MALOnt, an ontology for malware threat intelligence, for structured information extraction and knowledge graph creation. These processes are essential for cyber threat analysis, detection, classification, and attribution. Khan et al. [132] presented the dictionary template and RL for unstructured threat intelligence extraction for enhancing entity recognition in threat intelligence reports using DT-BERT-BiLSTM-CRF. They improved the efficiency of relationship extraction using NR-RL-PCNN-ATT, showing the power of AI in processing vast volumes of cybersecurity data. Yang and Lam [101] proposed a framework for automatically classifying significant volumes of CTI reports into SOCs. Based on document embedding models and ML classifiers, their approach clustered similar threat reports and enhanced the effectiveness of CTI analysis by an order of magnitude. This enhanced the effectiveness of early warning systems in devising proactive countermeasures against potential cyberattacks.

Schlette et al. [102] developed the CTI-SOC2M2 model for wide cybersecurity operation; it is a capability maturity model, in which CTI integration is a proxy for SOC service maturity. This model accepts the systematically developed approach of maturity models to enhance intelligence-driven security operations; however, its practical applicability relies solely on the data quality and availability of CTI. Kaiser et al. [103] studied cyber threat attribution and suggested an approach to automate the extraction of traits from unstructured reports of CTI using NLP and ML. Their newer embedding model, “Attack2vec,” considerably improved the detection accuracy at the attribution of cyber threat actors and outperformed the existing state-of-the-art models. Hettema [99] presented the minimum viable design of an SOC for a modern grid environment using tools such as Zeek and MISP, which enabled real-time threat intelligence sharing and incident response capabilities. This design reduced false positives and provided contextual information about alerts, thereby accelerating the incident handling process.

Tundis et al. [104] proposed a classification model for CTI in the organizations from South Africa by stating the importance of using CTI to prevent cyber threats. Alharbi [35] qualitatively studied SOCs and concluded that very high false-positive rates, low quality of threat intelligence, and poor automation are significant challenges. Moreover, effectively maintaining the performance of SOCs is challenging, particularly in the face of advanced cyber-attacks. They further emphasized that management and SOC staff share dif-

ferent opinions, which may impact the effectiveness of SOC operations. Muhammad et al. [105] elaborated on incident response automated systems using CTI to increase the speed and accuracy of defensive measures. Using a combination of datasets from several MITRE frameworks, their approach partially nullifies the first-mover advantage of attackers and ensures that automated systems are accurate and adaptive to different attack modes.

AI and ML technologies play a significant role in enhancing cyberthreat intelligence (CTI) capabilities, enabling improved detection and response to security incidents. However, challenges remain, such as the computational complexity of deep-learning models and the need for tailored solutions that address specific organizational threats without overlooking broader risks. While frameworks utilizing AI have shown promise in automating threat detection and improving operational efficiency, dependency on high-quality data and the balance between automation and accuracy are ongoing concerns. Moreover, the emphasis on a human-centric approach in cybersecurity illustrates the necessity for analysts to make threat information comprehensible and actionable. Collaborative platforms for information sharing have also emerged as essential for enhancing collective defense against cyber threats; however, their effectiveness depends on the willingness of organizations to share sensitive data. Additionally, the application of generative AI and natural language processing (NLP) for threat analysis offers the potential for deeper insights; however, challenges related to explainability, and trust must be addressed.

D. APT

In this dynamic and constantly changing field of cybersecurity, APTs have worsened with rapidly changing field of cybersecurity. Fig. 8 provides example for APT attack. New methods have been proposed for detecting, analyzing, and mitigating such threats. Rass et al. [3] proposed a game theory approach toward APT defense, highlighting that attack graphs play an important role in devising optimal defensive strategies. The model was built on previously developed models by including realistic attack actions, going beyond simple Poisson distributions. Results revealed that their proactive defense approach outperformed simple heuristic methods. However, limitation of their study was that they assumed the attacker already penetrated the system, which cannot be the case in all practical scenarios. The game theory settings considerably enhanced defensive techniques; however, the results have to be further validated for applying this model to different fields. Wang et al. [13] addressed the problem of alarm fatigue—a critical issue in SOCs filled with massive alarms. Their proposed solution, AlertPro, harnessed RL of triage alarms based on contextual and historical features, considerably improving the efficiency of the proposed process AlertPro could detect previously unnoticed attacks within available public datasets, thus improving security posture. However, this approach was based on partial feedback; better real-life results could be enriched if enriched.

Wang et al. [106] proposed an approach that reduced the alert volume via a reduction process and rebuilt APTs with high confidence using Monte Carlo tree search. Evaluations on a modified dataset yielded high performance concerning false negatives, providing significance of the tool in the long run for successful APT detection. However, further research is needed to determine the effectiveness of this method in realistic setups featuring different network configurations.

Nguyen et al. [107] detected abnormal DNS traffic, a common tactic used by APTs to evade traditional security systems. Their comparative analysis of unsupervised ML algorithms revealed that combining DBSCAN and KNN achieved a 100% detection rate with a low false-positive rate. This methodology provided a possible solution for identifying malicious DNS traffic; however, they admitted the need for further enhancements to deal with more complex attack scenarios and larger datasets.

Benzekri et al. [108] introduced a situation-driven security management aspect using their proposed framework called dynSMAUG. This approach applied situation awareness to manage dynamism in security systems, easing the identification of security and extraction from massive log data. The applied exploratory analysis techniques contributed to extending the dynSMAUG solution and effectively detected complex cyber-attacks such as APTs. They also indicated that one of the barriers to the full exploitation of the data available is related to the skills of security administrators in data science. Ren et al. [109] proposed CSKG4APT, a cybersecurity knowledge graph for APT organization attribution. CSKG4APT integrates a large quantity of fragmented intelligence, enabling proactive defense strategies rather than conventional passive methods. It also highlights the effectiveness of knowledge graphs by correctly attributing APTs and modifying the defense strategy accordingly. Although significantly developed, the accuracy of knowledge graphs should be continuously updated and enhanced to meet continued developments in deep learning algorithms. Presekal et al. [140] proposed a method that integrates semi-supervised deep packet inspection, traffic dispersion graphs, and the Enhanced Graph Convolutional Long Short-Term Memory (EGC-LSTM) model to address anomaly detection in Cyber-Physical Power Systems (CPPS). This AI-based approach detects and predicts anomalies by employing a resilient associative method to identify zero-day attacks. The evaluation demonstrated that the EGC-LSTM model outperformed existing spatio-temporal deep-learning models, achieving the lowest mean square error. The proposed method resulted in three key outcomes: spatio-temporal Advanced Persistent Threat (APT) detection, correlation, sequence prediction, identification of zero-day attacks, and detection of circuit breakers opened owing to cyber-attacks. However, the authors acknowledged a significant limitation in their approach, as the methodology did not consider insider threats, which differ from external threats. This gap indicates that adapting to these challenges is a potential area for future research.

APT have become increasingly sophisticated, requiring the development of innovative detection, analysis, and mitigation strategies. Recent approaches have highlighted the importance of utilizing advanced methodologies, such as game theory, to create optimal defense strategies, although some models assume prior system penetration, which may not always reflect real-world scenarios. Reinforcement learning techniques can tackle alarm fatigue in SOC by triaging alerts based on contextual data. However, the need for improved feedback mechanisms remains. Additionally, the integration of unsupervised machine learning methods for detecting abnormal DNS traffic has shown promising results, achieving high detection rates, while acknowledging the necessity for further refinement in complex environments. The introduction of frameworks emphasizes the role of situational awareness in managing security systems, although the skill gap among security administrators in data science poses a challenge. Moreover, the development of cybersecurity knowledge graphs facilitates proactive defense strategies by improving the attribution of APT, yet the accuracy of these graphs must be continuously updated to keep pace with advancements in deep learning. Incorporating semi-supervised deep packet inspection and enhanced machine learning models has demonstrated the ability to predict and detect anomalies, including zero-day attacks, although limitations in addressing insider threats indicate the need to focus on comprehensive security strategies that encompass both external and internal risks.

E. BLOCKCHAIN IN SOC

Integrating blockchain technology into a large field of cybersecurity applications will be duly regarded as a highly significant innovation in security to guard against the ever-changing realm of cyber threats. Fig. 9 shows how blockchain can be used in SOC. Recent studies have pointed out that blockchain can enhance security operations, specifically for lessening DDoS attacks, intrusion into autonomous vehicles, and disseminating intelligence on cyber threats. Based on the research of Yeh et al. [110], SOChain is a privacy-centric information-sharing service specifically for distributed denial-of-service. This solution allows safe and fair time sharing of DDoS data among distributed SOC with the help of blockchain technology and an incentive model based on the DDoS_coin token. This decentralized approach resolves the problems usually associated with trust to become a bottleneck to sharing data among the SOC. A significant improvement offered by SOChain is the introduction of a two-layer Bloom filter, which increases privacy by enabling effective searches while protecting sensitive information. However, despite its benefits, blockchain and smart contracts make the platform challenging regarding possible scalability and the need for a foolproof content verification system to ensure information integrity. Alaeifar et al. [111] discuss various blockchain-based architectures for CTI sharing, noting that while blockchain can enhance trust and security,

TABLE 3. Overview Table of SIEM and IDS in SOC.

Title	Tool	Authors	Problem Addressed	Solution Proposed	Limitations
Threat Classification Model for Security Information Event Management Focusing on Model Efficiency	SIEM	[62]	Overhead and inefficiency in deep learning-based threat classification	Developed CNN-static(1D) to reduce overhead while maintaining accuracy	It may still require significant computational resources
Threat Hunting as Cyber Security Baseline in the Next-Generation Security Operations Center	SIEM	[63]	Difficulty in managing multi-million-dollar security infrastructures	Integration of SIEM with Threat Hunting Techniques	Requires skilled analysts and may be cost-intensive
Sonification to Support the Monitoring Tasks of Security Operations Centres	SIEM	[64]	Challenges in network security monitoring tasks in SOC's	Used sonification to enhance SIEM tools, improving task performance	Needs validation in real-world SOC environments
The Challenge of Detecting Sophisticated Attacks	SIEM	[17]	Difficulty in detecting sophisticated multi-stage cyber-attacks	Developed novel features for SIEM tools based on expert SOC input	Limited by reliance on expert knowledge for feature development
SOCaaS-IoT: A Security Operations Center as a Service Approach for IoT Applications Using Open-Source SIEM	SIEM	[65]	Challenges in securing IoT networks	Proposed SOCaaS using open-source SIEM for IoT security	Requires robust integration with existing IoT networks
SIEM Architecture for the Internet of Things and Smart City	SIEM	[66]	Security threats in smart cities and IoT	Proposed SIEM-based architecture for monitoring IoT devices in smart cities	High scalability requirements; implementation in real-world scenarios untested
The Applicability of a SIEM Solution: Requirements and Evaluation	SIEM	[67]	Difficulties in selecting appropriate SIEM solutions	Proposed a framework for evaluating and selecting SIEM solutions	Limited by the accuracy of the evaluation criteria
MalRank: A Measure of Maliciousness in SIEM-based Knowledge Graphs	SIEM	[68]	Identifying malicious entities in SIEM environments	Developed MalRank, a graph-based algorithm to score maliciousness	May require extensive data and computing resources for real-time analysis
New Approach for Threat Classification and Security Risk Estimations Based on Security Event Management	SIEM	[69]	Inadequate threat classification and risk estimation in SIEM systems	Proposed a new model for classifying threats and estimating security risks	Implementation challenges in real-world environments
Towards trustworthy cybersecurity operations using Bayesian Deep Learning to improve uncertainty quantification of anomaly detection	IDS	[25]	Uncertainty in anomaly detection results leading to false positives	Bayesian Autoencoder (BAE) model with heteroscedastic aleatoric uncertainty	Computational overhead in real-time applications
VISNU: A Novel Visualization Methodology of Security Events Optimized for a Centralized SOC	IDS	[26]	Difficulty in real-time analysis of numerous IDS alerts in centralized SOC's	VISNU, a visualization system for classifying and displaying security events	Limited visualization may not improve detection accuracy.
Transformer-based framework for alert aggregation and attack prediction in a multi-stage attack	IDS	[28]	Overwhelming volume of alerts and lack of effective prediction models	Transformer-based model with similarity-based alert aggregation	Requires extensive computational resources and data
Stream clustering guided supervised learning for classifying NIDS alerts	IDS	[30]	High volume of low-importance or false-positive NIDS alerts	Stream clustering combined with supervised learning for alert classification	Limited by the quality of the initial clustering process
Off-policy actor-critic deep reinforcement learning methods for alert prioritization in intrusion detection systems	IDS	[32]	Instability and limited exploration in DRL-based alert prioritization	TD3-AP and SAC-AP, off-policy DRL methods for alert prioritization	Complexity in implementation and tuning
On using contextual correlation to detect multi-stage cyber-attacks in smart grids	IDS	[29]	Difficulty in detecting coordinated, multi-stage cyber-attacks in intelligent grids	Contextual correlation approach using cyber intelligence from multiple sources	Challenges in real-time application and scalability
Methods for Host-Based Intrusion Detection with Deep Learning	IDS	[33]	Challenges in detecting novel attacks with HIDS	ALAD algorithm using deep learning for sequence-to-sequence behavior in system calls	Black-box nature of deep learning models
HH-NIDS: Heterogeneous Hardware-Based Network Intrusion Detection Framework for IoT Security	IDS	[34]	High computational cost of AI-based IDS for IoT	Lightweight ANN models deployed on high-performance, low-power hardware	Hardware dependence may require specialized devices
Reduction of Alert Fatigue using Extended Isolation Forest	IDS	[11]	Overwhelming volume of alerts leading to alert fatigue	Extended Isolation Forest for identifying anomalous alerts	May miss low-frequency, high-impact alerts
Towards trustworthy cybersecurity operations using Bayesian Deep Learning to improve uncertainty quantification of anomaly detection	IDS	[25]	Uncertainty in anomaly detection results leading to false positives	Bayesian Autoencoder (BAE) model with heteroscedastic aleatoric uncertainty	Computational overhead in real-time applications

the current landscape remains fragmented and inadequately explored. The paper suggests future research directions, particularly in developing standardized protocols and addressing the legal complexities of cross-border data sharing.

Various blockchain-based architectures for CTI sharing, noting that while blockchain can enhance trust and security, the current landscape remains fragmented and inadequately explored. The paper suggests future research directions, particularly in developing standardized protocols and addressing the legal complexities of cross-border data sharing.

The limitations of blockchain under this context include its ability to worsen interoperability issues if not correctly standardized, as well as the challenges to guarantee data accuracy and integrity in a decentralized architecture.

Another study involved enhancing trust and security in AVs employing improvisations through blockchain technology in Industry 5.0 by Anbalagan et al. [112]. The authors proposed a Blockchain-assisted Intrusion Detection System (BIDS) empowered by machine learning, particularly Stochastic Gradient Descent (SGD), for intruder detection in the proposed IoV environment. Blockchain integration provides a secure and trustworthy environment for data exchange among AVs, addressing the current data privacy problem. The results obtained from this work show that BIDS achieved a high accuracy of 98%, outperforming the current state-of-the-art solutions. Execution of such a system in real-world AV networks, however, would be hampered by the computational overhead of blockchain and machine learning processes, as well as a need for widespread adoption of blockchain infrastructure in the automotive industry. In response to such challenges, Graf and King [113] propose the following narrower application of blockchain technology to enhance awareness of cyber situations related to the protection of Critical Infrastructure. It presents a model combining deep autoencoder neural networks with blockchain to automate the classification and management of cyber incident reports. The blockchain allows for secure data sharing of incidents and manages information to reduce manual operations at the security operation center. Smart contracts automate workflows, thus enabling real-time incident management. Despite having the potential for better visualization, the approach also faces issues of AI and blockchain technology integration, like high computational demand and the need for real-time processing, which might impose limitations on its applicability in large-scale CI environments. Table 4 provides a summary for the Blockchain solutions in SOC.

The integration of blockchain technology into cybersecurity applications represents a significant advancement in safeguarding against the evolving landscape of cyber threats. Blockchain can enhance security operations by addressing issues such as Distributed Denial-of-Service (DDoS) attacks and improving the sharing of cyber threat intelligence (CTI). However, challenges such as scalability concerns and the need for effective content verification to ensure data integrity remain. Furthermore, innovations combining deep learning with blockchain aim to automate cyber incident management,

although these approaches also face challenges of integration and high computational demands.

F. AUTOMATION IN SOC

Automated and effective response modes have become crucial in the current cybersecurity domain due to rapidly changing attacks. Fig. 10 shows how SOAR is used to automate SIEKM actions. Several studies have suggested ways to enhance the automation and effectiveness of the response, highlighting new perspectives and pertinent shortcomings. Hughes et al. [114] reported an illustrative approach, which focused on a model-free RL technique for intrusion response systems. Such tasks present challenges resulting from dynamical network topology characteristics, which may impair current effective non-model-based methods. Using a RL agent trained in a simulated environment, complex multistage attacks can be prevented while minimizing the interference of legitimate activities within the network. However, the limitations of their approach was that the performance of the approach was strictly limited by the quality and completeness of training, which might only cover some of the possible attack variations. Coscia et al. [115] addressed the challenges of detecting and neutralizing DoS and DDoS attacks using a DT-based network intrusion detection and prevention system. They proposed an algorithm, Anomaly2Sign, to collect rules for Suricata in an unsupervised manner from both normal and abnormal traffic patterns. The decision tree model exhibited better classification performance, providing an accuracy of 99.7%–99.9% that surpassed those of other ML-based classifiers. These datasets cannot be regarded as representative data due to potential risk for overfitting, which may impact the practical use of the developed rule set. Kinyua and Awuah, [5] debated on the future of SOAR and the use of AI and ML for empowering SOC analysts. Their study suggested that SOAR solutions integrated with SIEM systems can enhance the automation and efficiency of security operations. Islam et al. [116] proposed a declarative API-driven orchestration (DecOr) framework for SOAR platforms to address the challenges of maintaining orchestration due to constant changes in incident response process and security toolchains. DecOr eases orchestration by abstracting complexities in security operations using declarative APIs; therefore, one can respond more effectively and correctly. They also highlighted that the experiments conducted using their proposal yielded higher precision and recall values. However, the authors highlighted that the effectiveness of their framework was based on the accuracy of the ontological knowledge base. Sworna et al. [4] proposed APIRO, an architecture for automatic recommendation of APIs for security tools in SOAR solutions. Within the context of data scarcity, diversity, and semantic differences, APIRO used ML methods and performed data augmentation within the convolutional neural network architecture. Results showed significant improvement in the precision of API recommendations, with Top-1 accuracy of 91.9%. However, the

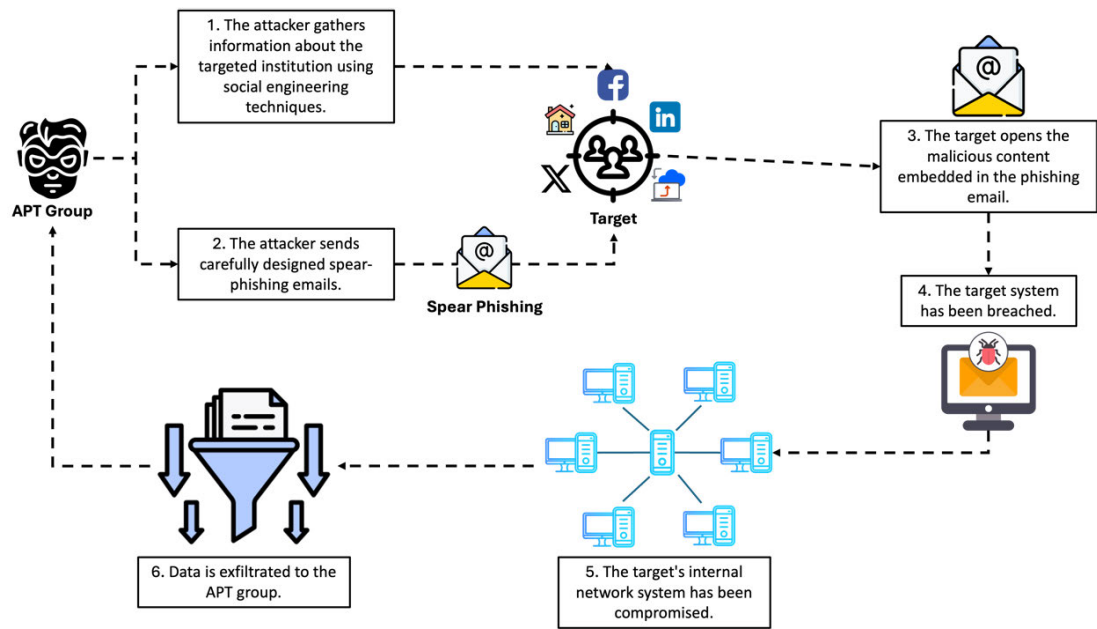


FIGURE 8. APT attack example.

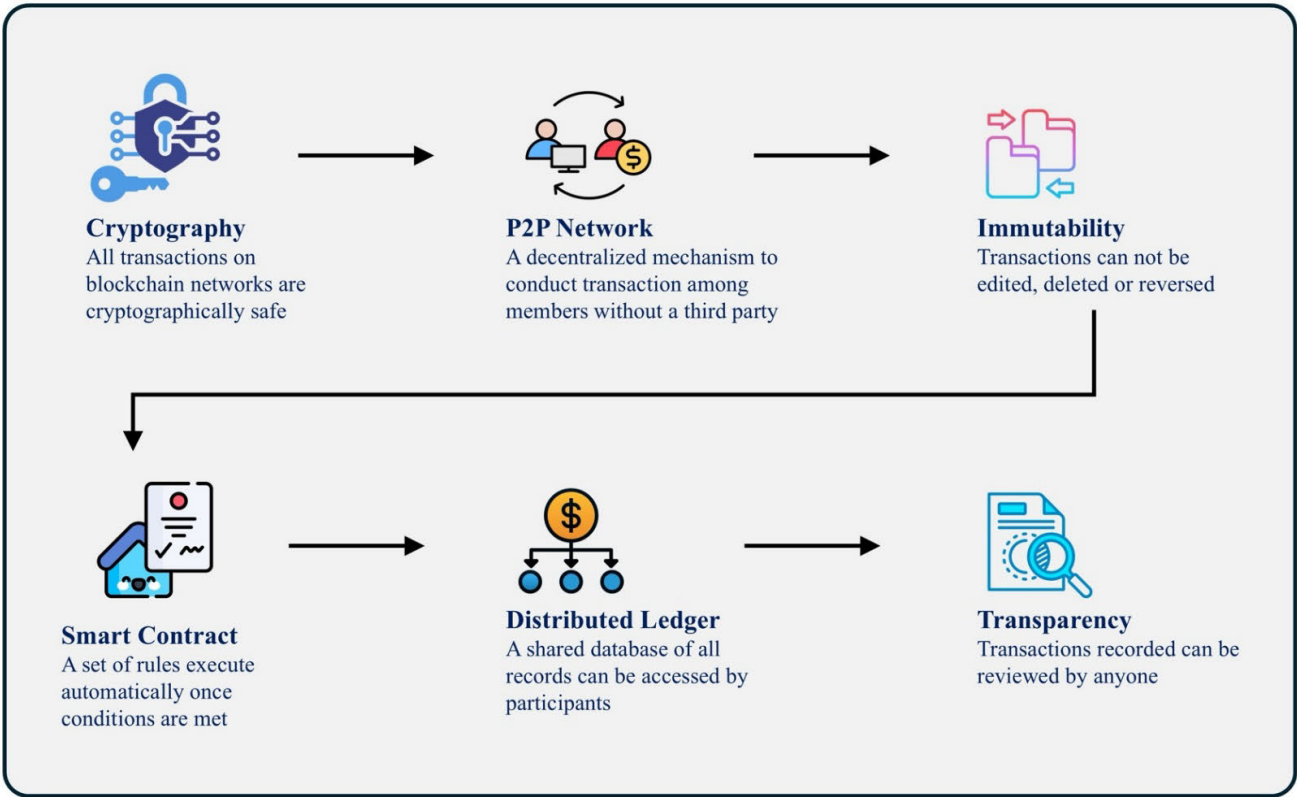


FIGURE 9. Blockchain in SOC.

effectiveness of APIRO was highly dependent on the quality of training data and the diversity of security tools used during the review. Sridharan and Kanchana [9] integrated

the SIEM system with SOAR platform to analyze automation based on the current complex cyber threat landscape as well as the weaknesses inherent in reliance on traditional

TABLE 4. Summary Table of Blockchain.

Title	Authors	Problem Addressed	Solution Proposed	Limitations
SOChain: A Privacy-Preserving DDoS Data Exchange Service Over SOC Consortium Blockchain	[110]	Lack of trust and fairness in DDoS data sharing among SOCs	Proposed SOChain, a decentralized DDoS data exchange platform using blockchain and dual-level Bloom filters	Limited by the need for widespread adoption among SOCs, scalability, and privacy concerns in large networks need further exploration
Blockchain-Assisted Hybrid Intrusion Detection System in Autonomous Vehicles for Industry 5.0	[112]	Vulnerability of autonomous vehicles to cyber infiltration due to lack of trusted data exchange	Proposed a Blockchain-assisted Hybrid IDS using Machine Learning for trust evaluation in 5G-V2X IoV environments	Requires high computational resources; effectiveness in real-world deployment across various IoV networks needs validation
Current Approaches and Future Directions for Cyber Threat Intelligence Sharing: A Survey	[111]	Challenges in CTI sharing due to legal, interoperability, and data reliability issues	Conducted a comprehensive survey of current CTI-sharing approaches and proposed future research directions	Theoretical in nature, practical solutions for overcoming identified challenges need development and validation.
Neural Network and Blockchain-Based Technique for Cyber Threat Intelligence and Situational Awareness	[113]	Need for efficient and secure cyber incident analysis and situational awareness.	Proposed a neural network and blockchain-based approach for automated incident classification and management	Limited by the initial setup complexity, effectiveness in large-scale, real-time scenarios needs further testing.

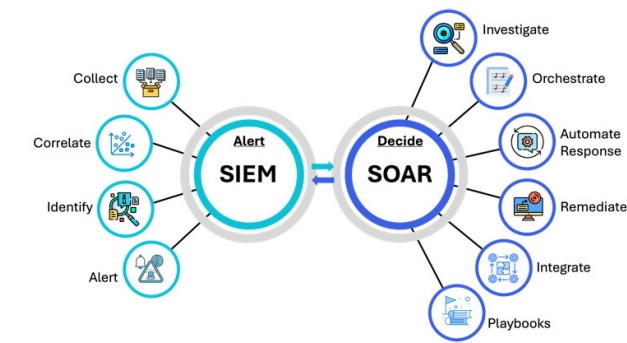


FIGURE 10. SIEM and SOAR integrations.

security tools such as firewalls and antivirus software. They also indicated that integrating SOAR with SOC provides an organization with tremendous opportunities to reduce manual workloads and response times. However, its integration with SOAR introduced some challenges, particularly regarding the flexibility of automation processes in changing cyber threat landscapes. Abeywardena et al. [117] presented an approach to automate threat advisory report generation using threat intelligence. They emphasized that although manual generation of reports is time-consuming, they must be tailored for specific organizational needs. By automating this process, SOC teams can respond more swiftly to incidents. However, the quality of automated reports depends heavily on the accuracy and relevance of the threat intelligence data, which can vary considerably across different sources. Islam et al. [118] analyzed architectural underpinnings required to build effective SOAR platforms and proposed a conceptual model and critical design considerations that guided security tool integration and automation of incident response processes. Although their preliminary evaluations reported the efficacy

of the proposed architecture, it is highly sensitive in reality to the specific requirements of SOCs. Thus, ensuring the flexibility of design is crucial. Table 5 provides a summary for automation solutions in SOC.

Various studies have explored ways to enhance automation in incident response, revealing both innovative strategies and their inherent limitations. For instance, the use of reinforcement learning (RL) agents can effectively mitigate complex multistage attacks, although their performance relies heavily on the training quality and comprehensiveness. Additionally, decision-tree-based algorithms have demonstrated high classification accuracy for detecting denial-of-service attacks, but concerns about overfitting highlight the need for representative training datasets. The integration of AI and machine learning with SOAR can empower security analysts and streamline operations; however, challenges remain in maintaining orchestration because of the evolving nature of security processes. Frameworks like DecOr aim to simplify orchestration complexities, yet their effectiveness is contingent on the accuracy of underlying knowledge bases. Furthermore, architectures such as APIRO improve the recommendation of APIs for security tools, although their success depends on the quality of the training data. Although integrating SOAR with Security Information and Event Management (SIEM) systems opens opportunities for reducing manual tasks and response times, it also introduces challenges related to automation flexibility in a constantly changing threat landscape. Automating threat advisory report generation can enhance response times for SOC teams, but the quality of these reports is contingent on the reliability of the threat intelligence used. Overall, the insights underline the importance of adaptability, data quality, and comprehensive training in developing effective automated response systems within cybersecurity operations.

TABLE 5. Summary Table of automation.

Title	Authors	Problem Addressed	Solution Proposed	Limitations
Design and Generation of a Set of Declarative APIs for Security Orchestration	[116]	Difficulty in adapting to changes in SOAR platforms due to complex architecture	Proposed DecOr, a Declarative API-driven orchestration approach to hide complexity and adapt to changes	It may require extensive implementation effort and validation in diverse environments.
APIRO: A Framework for Automated Security Tools API Recommendation	[4]	Challenges in manual integration of security tools in SOAR platforms	Proposed APIRO, a learning-based framework for automated API recommendation using CNN and data augmentation	Limited by the scope of security tools tested, performance may vary across different SOC environments
Dynamic Countermeasure Knowledge for Intrusion Response Systems	[114]	Lack of dynamic countermeasure sharing in Intrusion Response Systems	Proposed a countermeasure standard for automated sharing and adoption of countermeasures	Effectiveness in real-world SOC needs validation; possible overkill countermeasures if misapplied
A Model-Free Approach to Intrusion Response Systems	[10]	Need for effective automatic response to alerts in increasingly dynamic networks.	Proposed a model-free reinforcement learning approach for IRS in a virtualized testbed	Limited to specific scenarios tested, scalability, and generalization to more complex networks need validation.
Automatic Decision Tree-Based NIDPS Ruleset Generation for DoS/DDoS Attacks	[115]	Lack of robust and evolving NIDPS rulesets for DoS/DDoS detection	Proposed Anomaly2Sign, an algorithm to generate NIDPS rules using Decision Trees, achieving high detection rates	Performance may vary across datasets; it requires constant updating to adapt to new threats
SIEM Integration with SOAR	[9]	Advanced cyber threats bypass traditional perimeter controls, requiring SOAR integration.	Proposed integration of SIEM with SOAR to reduce alert volume and improve response times	Effectiveness in diverse organizational contexts needs further exploration.
Automating Threat Advisory Report Generation Using Threat Intelligence	[117]	Manual creation of threat advisory reports is time-consuming and prone to errors.	Proposed an automated framework for generating threat advisory reports using threat intelligence	It may require customization for specific organizational needs, as the scope of threat databases and RSS feeds used limits it.

G. SOC EFFICIENCY AND INTEGRATION

The incorporation and functioning of SOC in contemporary cybersecurity environments require a diverse strategy, as many studies have proven to underscore the advancements and constraints in this domain. Ofte and Katsikas [16] extensively analyzed situational awareness in SOC, which is essential for effective cybersecurity administration. Their systematic review identified the reliance on Endsley's three-level situation awareness model that extended into the domain-specific model of cyber situation awareness. However, they identified a significant gap in the literature: many studies need a solid theoretical foundation, particularly those developing tools to enhance situational awareness. This indicates the need for a broadened use of system-based perspectives such as distributed situational awareness to bridge human cognitive processes with complex SOC environments. Efficient management and validation of cyber threat data are some challenges in SOC, and Islam et al. [138] proposed the SmartValidator framework to address these challenges. This AI-based framework automated the validation of alerts using ML techniques to limit manual intervention in rule updates. Their evaluation showed that dynamic model construction was much more efficient and required 99% fewer models than traditional methods while maintaining high accuracy (F1-score greater than 0.8 for 75% of the models). However, they also acknowledged that while the framework increased efficiency, its efficacy depended on the quality of input data and the dynamic needs of SOC. Therefore, adapting to rapidly changing contexts might be a limitation. Another innovation is the integration of virtual reality (VR) into cybersecurity operations. Munsinger et al. [31] developed a VR tool that

enhanced the situational awareness of SOC operators, particularly novices. The initial results of their study are promising; however, the small sample size limited the generalizability of findings, necessitating further research to validate the effectiveness of the VR tool in more extensive and diverse settings.

Building upon the need for automation within SOC, Sworna et al. [119] introduced IRP2API, a framework to automate the mapping between IRPs and APIs of security tools. This framework applied unsupervised transfer learning to reduce the amount of manual effort in API mapping and achieved a Top 15 accuracy of 91.1%. Although very effective in practice, the framework relies on pretrained embedding methods that may not generalize well to new or specialized SOC settings, where never-before-seen or changing threats rapidly require updated models. Onwubiko and Ouazzane [120] focused on a rarely discussed topic, cyber onboarding for SOC. They discussed several technical and organizational challenges in inserting a new system into an SOC monitoring framework. They highlighted significant challenges such as complexities in establishing systems for generating appropriate logs and hurdles in aligning human workflow with technical specifications. These pitfalls underscored the need for a more streamlined onboarding process that can better meet diverse needs from technical staff to executive management via reduced operational inefficiencies. Mutemwa et al. [121] further discussed the issues associated with the integration of SOC, particularly during their deployment within existing IT systems, current procedures, and policies, including ticket logging and reporting mechanisms. They prioritized the systems that required monitoring and proposed aligning SOC operations with organizational

TABLE 6. Overview of AI/ML proposed solutions in literature.

Title	Ref	Tool	Dataset	Algorithm	Purpose	Accuracy	Precision	Recall	Real-Time
Threat Classification Model for Security Information Event Management	[62]	SIEM	2.6 million threat events (SOC data)	CNN-static(2D), LSTM, GRU	Efficient deep learning model for threat classification in SIEM	94.07%-95.11%	High	Highest	Yes
Design and Generation of a Set of Declarative APIs for Security Orchestration	[116]	Automation	Real-world playbooks	Declarative APIs (DAPL), NLP	Automate and simplify SOAR orchestration by using declarative APIs	93%	>80%	>80%	Yes
MalRank: A Measure of Maliciousness in SIEM-based Knowledge Graphs	[68]	SIEM	3TB of SIEM data	MalRank, Graph-based inference	Infer maliciousness scores for entities in SIEM-based knowledge graphs	AUC = 96%	High	High	Yes
Blockchain Assisted Hybrid Intrusion Detection System in Autonomous Vehicles for Industry 5.0	[112]	Blockchain	5G-V2X Internet of Vehicles (IoV)	Stochastic Gradient Descent (SGD), Blockchain	Enhance IDS in Industry 5.0 with blockchain for secure AV data exchange	98%	Not specified	Not specified	Yes
Detecting Abnormal DNS Traffic Using Unsupervised Machine Learning	[107]	APT	Boss of the SOC Dataset	DBSCAN, KNN	Detect malicious DNS traffic to identify potential APT activities	100% detection rate	Not specified	1.6%–2.3% false-positive rate	Yes
HH-NIDS: Heterogeneous Hardware-Based Network Intrusion Detection Framework for IoT Security	[34]	IDS	UNSW-NB15, IoT-23	Lightweight Neural Network	High-performance, low-power anomaly detection in IoT networks	98.57%, 99.66%	N/A	N/A	Yes
APIRO: A Framework for Automated Security Tools API Recommendation	[4]	Automation	Multiple security tools' APIs	CNN, API-specific word embedding	Automate the recommendation of security tool APIs in SOAR environments	91.9% (Top-1 Accuracy)	91.9% (Top-1)	Not specified	Yes

workflows. However, they noted a general lack of coordination between technical and non-technical personnel during integration, which is a limitation that detracted from the overall effectiveness of SOC. Agyepong et al. [122] addressed performance measurement issues with SOC analysts by proposing the SOC analyst assessment method (SOC-AAM), which provided a structured approach to assess the performance of analysts based on key performance indicators (KPIs) in terms of the quality of incident analysis and reporting. This method was validated with an empirical evaluation in two SOC and effectively demonstrated the creation of a structured assessment framework. However, the effectiveness of the method involved accurate and uniform application of KPIs, which may differ considerably for distinct SOC. Dietz et al. [74] explored the digital twins of the physical assets of an industrial setup with the capability of situational awareness of the number of attacks in progress and how they can simulate attacks. Using this approach, such systems can be better equipped to handle threats in industrial

control systems; however, the likelihood of added difficulty in incorporating them into current integrations is profound. Perera et al. [76] focused on using chatbots in SOC to expedite the investigation process and reduce the cognitive load on analysts. They proposed an intelligent SOC chatbot that automated incident ticketing and suggested appropriate investigation flows based on similar incidents from earlier times. This approach can help to standardize responses and minimize human errors; however, the efficiency of such tools depends on their ability to learn and adapt to new threats with time.

Liu et al. [123] addressed the challenge of alert fatigue in SOC by proposing Context2Vector that relies on context representation learning to enhance event triage processes. Combining dense event representations with sparse topic representations outperformed state-of-the-art methods in effectively identifying high-risk events, doubling the attacker recall rate in some cases. However, this method was bound by context extraction from multiple behavioral viewpoints;

therefore, its application can be challenging, particularly in scenarios where sources are heterogeneous or not standardized. Spring and Illari [14] followed the review on human decision-making regarding computer security incident analysis, emphasizing the steps of evidence gathering, examination, and documentation. Although the review highlighted the strengths of existing standards, it also identified gaps in guidance on task prioritization and result interpretation under time constraints. This limitation suggested a need for more explicit decision-making frameworks to better support analysts in high-pressure situations. János et al. [124] focused on the security concerns within SOC operations, emphasizing the challenges of evolving cyber threats. They reported on the primary roles of SOC, namely incident detection, investigation, and response, and offered solutions on enhancing the capabilities of SOC by deploying sophisticated situational awareness tools. The work underpinned the continuous adaptation and improvement of SOC strategies for preventing modern cyber threats.

Tsakalidis et al. [125] presented a cybercrime incident architecture to standardize the response mechanisms to cybercrime incidents using an innovative adaptive response policy. Although ambitious in its attempt to have a comprehensive approach to cybercrime management, this architecture required modifications due to the manual generation of ARP. Further automation in this regard might help increase the applicability of this framework in real-world scenarios. Chhetri et al. [126] discussed a human–AI teaming framework for alleviating alert fatigue in SOC and proposed a framework that provided dynamic decision-making to facilitate smooth transitions between automated and collaborative modes. Thus, AI must be embedded within SOC to extend human capabilities. However, this methodology relied heavily on the ongoing enhancement of AI algorithms and the flexibility of human operators, both of which may pose difficulties in swiftly evolving threat landscapes. Ganesan et al. [127] presented an optimization model that scheduled cybersecurity analysts to minimize risk in SOC. Their model was scalable and effective in managing analysts' resources, keeping the level of risk within acceptable bounds. However, their model might require careful tuning to a wide range of SOC environments due to the nonlinear relationship between analyst-to-sensor ratios and risk; hence, its general applicability may be limited. Rosso et al. [128] discussed the need for a tool to measure performances of SOC and proposed SAIBERSOC, which employed synthetic attack injections to assess SOC performance based on metrics such as detection accuracy and time-to-investigation. Rosso et al. [128] exhibited that SAIBERSOC could appropriately identify performance variations in SOC, presented evidence that it is a valuable performance investigation tool for security researchers and operators striving for performance optimization in SOC. Table 6 provides a summary for AI/ML solutions proposed in the literature.

Analysts often struggle with information overload and alert fatigue, making it difficult to prioritize genuine threats among

a large number of notifications. Efforts to enhance situational awareness, while promising, depend on the quality of the data they rely on, which can vary greatly in real-world environments. Moreover, the integration of new systems into existing workflows frequently encounters obstacles owing to a lack of coordination between technical and non-technical staff, creating a gap that can limit collaborative efforts. As SOC strive to adapt to the complexities of modern cyber threats, the need for streamlined processes, effective training, and supportive tools that empower analysts rather than overwhelm them has become increasingly vital.

VI. REFERENCE ARCHITECTURAL MODEL FOR SOC IMPLEMENTATION

In this section we discuss and answer **RQ2** by proposing a next-generation SOC powered by AI and ML, which provides a whole-domain defense by combining the power of advanced technologies with human expertise. The proposed architecture was designed on three core layers, namely data acquisition, security, and presentation, each playing an essential role in safeguarding the digital assets of an organization. Fig. 11 provides an overview about the proposed reference architecture model.

A. DATA ACQUISITION LAYER

The data acquisition layer is the base of the SOC and collects information from various sources within the organizational network. These include

- **Network Devices:** Routers, switches, and firewalls log and provide traffic data that offer an insight into network activities.
- **Endpoints:** Workstations, servers, and mobile devices that contribute to logs, application usage data, and security events.
- **Databases:** Logs from critical databases, usually those storing sensitive information, that are continuously monitored.
- **Applications:** On-premises and cloud enterprise application logs provide valuable information about user activities or possible security risks.
- **External Threat Feeds:** Data collected from external sources such as IoCs and threat intelligence are included in the mix, ensuring that the SOC remains updated and aware of new threats.

The data acquisition layer thus ensures that all relevant data are captured, revealing the organization's security environment, and creating a foundation for the rest of the layers.

B. SECURITY LAYER

The security layer is the action area, the heart of the SOC, where AI and ML analyze data to detect threats and actions in real time. This layer comprises the following parts:

- **Intrusion Detection and Response (IDR):** This module serves as the first line of defense and applies AI-driven techniques such as convolutional neural

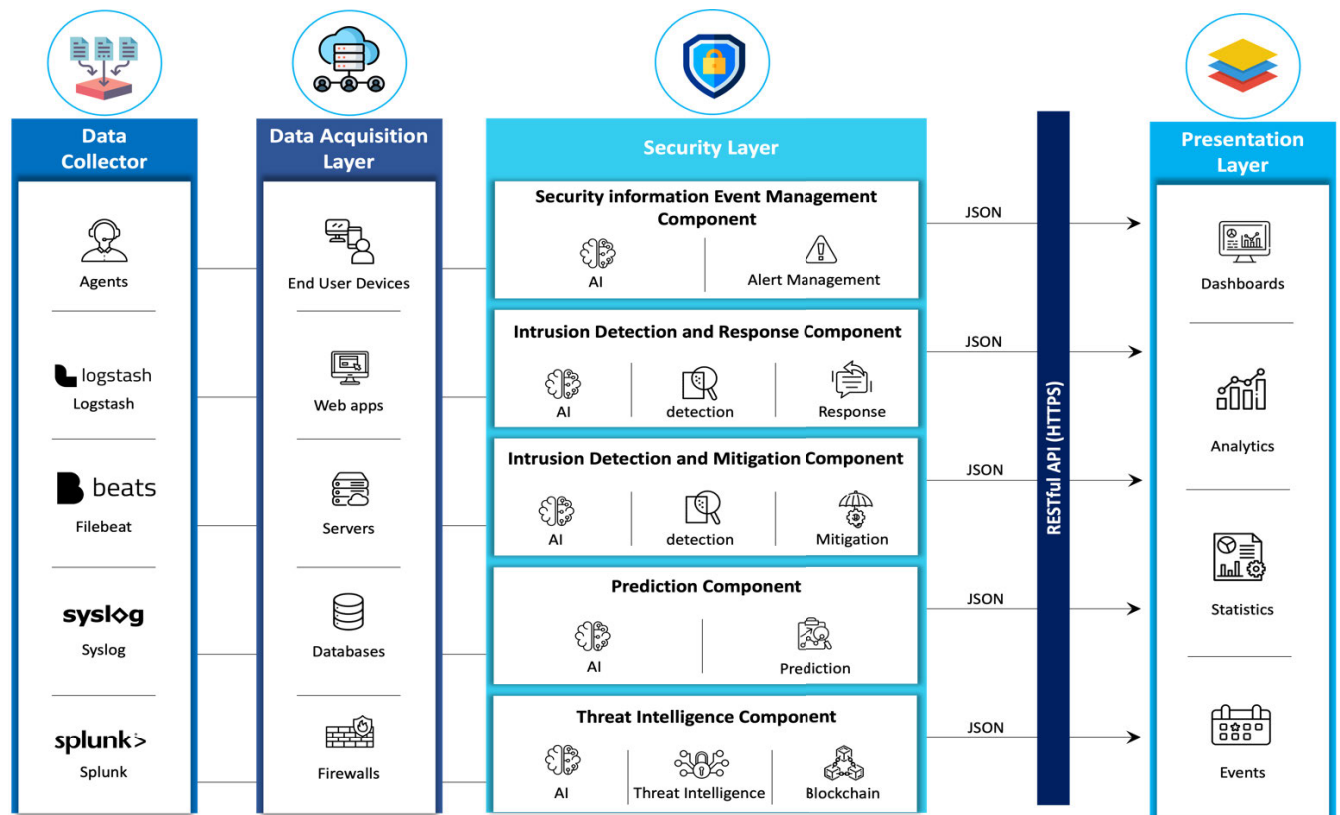


FIGURE 11. A comprehensive reference architectural model for NG SOC.

networks or recurrent neural networks to recognize patterns and identify anomalies. For instance, Akinrolabu et al. [17] reported that the system should be well-equipped to react to intrusion detection by isolating it and keeping the threat contained.

- **Intrusion Detection and Mitigation (IDM):** The IDM component, upon detecting a threat, exploits AI-driven techniques such as RL to confine and neutralize threats. GANs are used to generate attack scenarios to effectively prepare the mitigation strategy in advance [5].
- **Intrusion Prediction:** This module does not allow the attack to occur but leverages predictive analytics such as long short-term memory or autoencoder models to make predictions or profile a potential attack. Such models can analyze history and predict the occurrence of future attacks, thereby securing the SOC [28].
- **Threat Intelligence:** AI powers this component with algorithms such as graph neural networks to analyze and prioritize threat data from internal and external sources. This keeps the SOC ahead of all emerging threats and provides actionable insights [8].
- **SIEM System:** This is the central nervous system of the SOC. An AI-boosted SIEM system uses ML technologies to detect anomalies based on isolation forest or clustering by DBSCAN for prioritizing alerts and

reducing false positives so that the security expert can focus on the most critical incidents [62].

- **APT Detector:** APTs are demanding threats that require advanced detection techniques. An APT detector may use deep learning models based on an autoencoder or bidirectional LSTM networks to detect subtle signs that indicate an APT attack, allowing the SOC to intervene before the attack becomes considerable in size. In other words, the attack surface spirals out of control.

These components are not supposed to work in isolation; therefore, they work holistically to share information and intelligence that creates a single and adaptable security environment. For instance, threat intelligence may make data available and enhance the accuracy of intrusion prediction models. However, other parts such as the APT detector will provide critical information back into the SIEM system to increase the general visibility into threats.

C. PRESENTATION LAYER

The presentation layer enables analysts to oversee the technical operations of SOC. This layer aggregates all data and insights produced by the security layer, presenting it clearly and in an action-oriented manner. The key features of this layer are as follows:

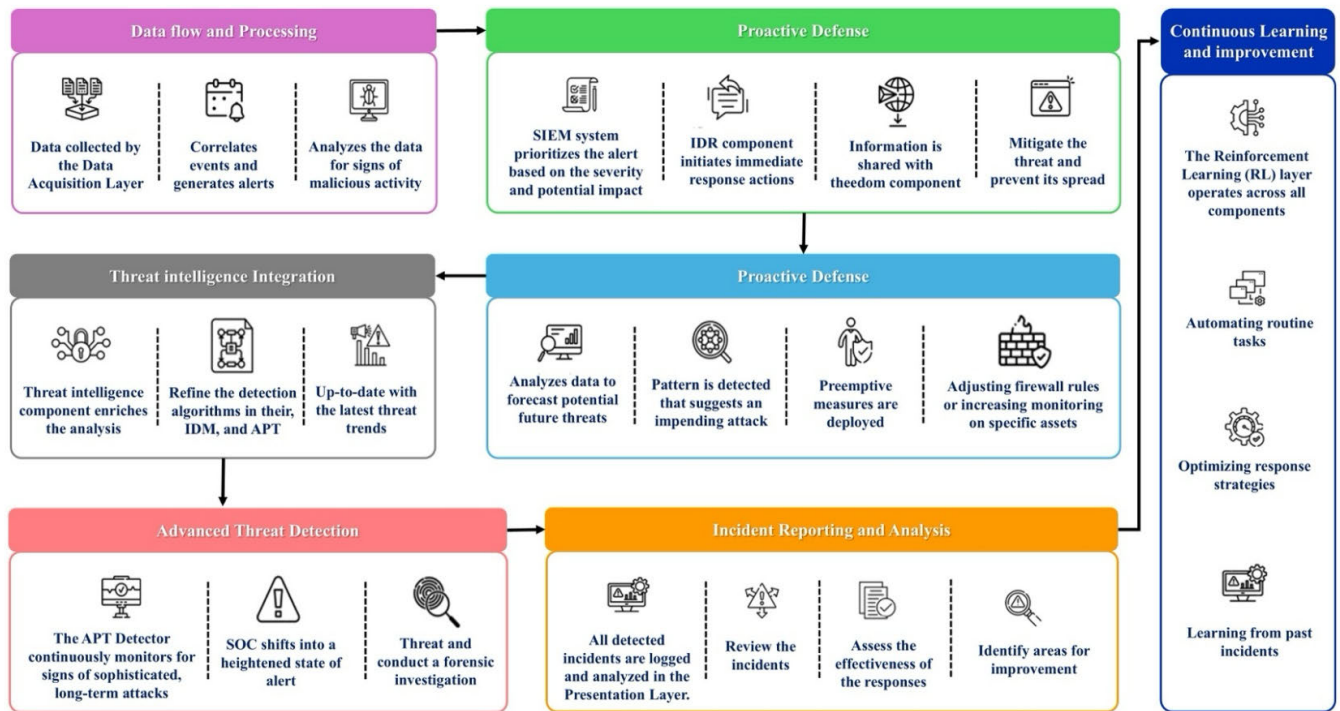


FIGURE 12. Workflow and components interactions.

- **Dashboard:** A system designed to provide an overview of the security situation in the organization at a particular time along with current metrics, alarms, and developments. AI can structure basic information displayed to users using RL.
- **Incident Analysis and Reporting:** Tools enable an analyst to review each event to analyze the security incidents, systems involved, and actions proposed for implementation. ML is employed to create reports that display potential trends and patterns to be examined further.
- **Threat Visualization:** Graphic displays illustrate the evolution and progress of identified threats in the network and measures taken to counter them. Methods such as dimensionality reduction (t-SNE) can reduce the complexity inherent in data, thus helping the analysts be more responsive to threats.

This layer eliminates any barriers to security analysts, helping them to obtain the required meaningful information at the right time.

D. WORKFLOW AND COMPONENT INTERACTION

The workflow in the proposed SOC begins with the data acquisition layer, where data are continuously collected from various sources. Fig. 12 shows the workflow and components interactions. The process unfolds as follows:

- **Data Flow and Processing:** The data acquisition layer collects data, which are ingested into the SIEM system. This layer then correlates events against alerts on those

events, whereas the intrusion detection and response component analyze that data to identify malicious activity indicators.

- **Detection and Response:** SIEM systems ranks the alert by severity and potential impact after an intrusion is detected. The IDR undertakes immediate response measures isolating affected systems and blocking malicious traffic. These measures are then passed onto the IDM component for taking further mitigative actions against threats and their propagation.
- **Proactive Defense:** The intrusion prediction component enables continuous data analysis to predict potential threats. Suppose a pattern that may relate to a future attack is detected, during which, measures such as adjusting the firewall rules or monitoring certain assets are implemented.
- **Threat Intelligence Integration:** The threat intelligence component enriches the context of analysis with external sources. This information will be helpful in re-calibrating detection algorithms in the IDR, IDM, and APT detection components to keep them updated with prevailing threats at any time.
- **Advanced Threat Detection:** The APT detector constantly monitors for any signs of advanced and long-term attacks. When it detects an APT, the SOC responds to high alert and all components work together to mitigate the threat and provide forensics investigation.
- **Incident Reporting and Analysis:** The presentation layer logs and analyzes all detected incidents and response actions. Security analysts review incidents,

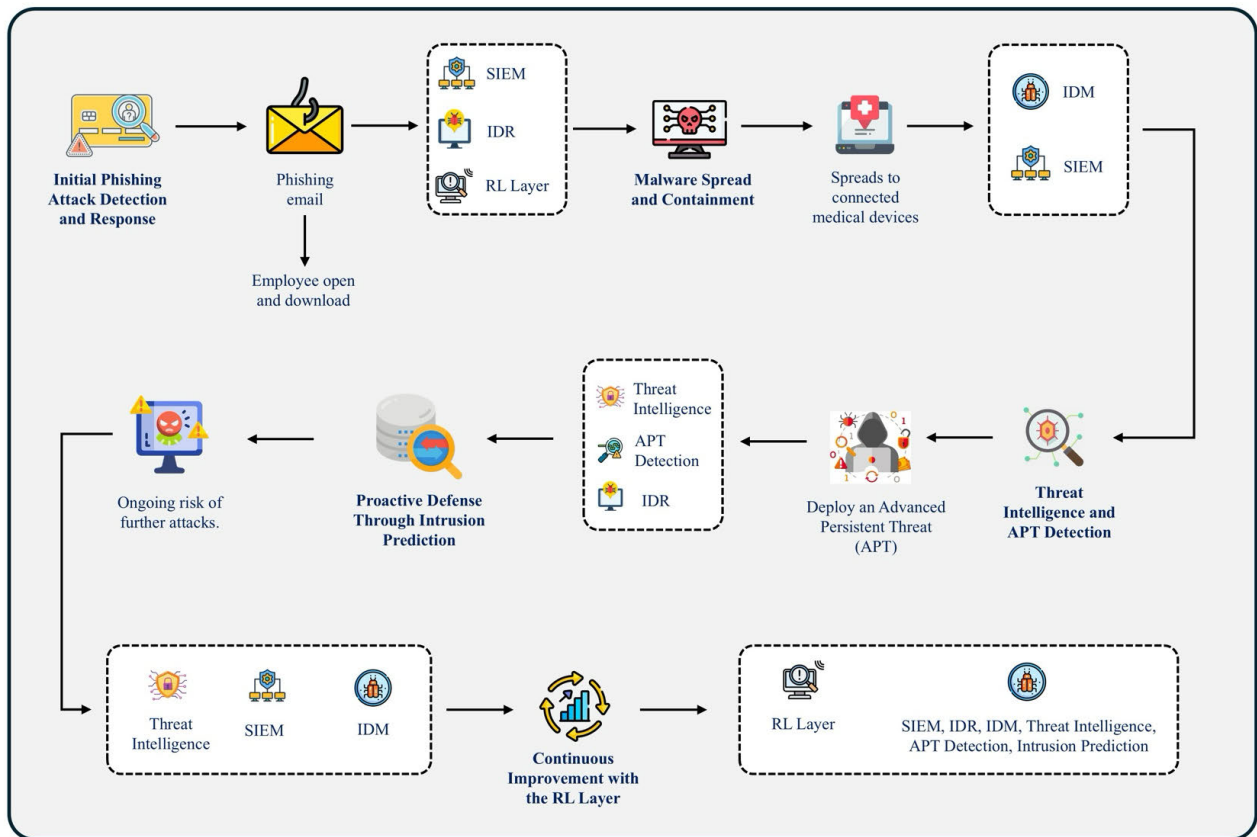


FIGURE 13. Use case scenario: Multistage cyber attack on healthcare institutions.

assess the effectiveness of their responses, and determine areas for improvement using the tools provided by this layer.

- **Continuous Learning and Improvement:** The RL layer spans each component, automates routine tasks, optimizes response strategies, and learns from past incidents. Such a continuous feedback loop would ensure that the SOC evolves and improves over time to adapt to new threats and operational efficiency refinements.

The proposed SOC infrastructure contains more than mere emerging technologies; it is a fully integrated system for detecting and responding to devastating cyber threats in real time. Using AI and ML at various layers, the SOC is empowered to deal with the modern complexities of cybersecurity. The interaction of different components, complemented by the ability of the RL layer to learn continuously, ensures that the SOC efficiently responds to current threats and looks out for the future. In this regard, security teams are better positioned to protect their institutions by offering them the necessary tools to outmaneuver cybercriminals and keep their most precious assets safe.

VII. USE CASE SCENARIO: A MULTI-STAGE CYBER ATTACK ON A HEALTHCARE INSTITUTION

In this use case, we aim to address **RQ3** by describing how a next-generation SOC powered by AI and ML defends a

healthcare institution from a sophisticated multi-stage cyber-attack. This also illustrates the role of each component in the SOC and how these components are integrated to provide security throughout.

A. SCENARIO BACKGROUND

A sophisticated, a multistage cyber-attack is performed on a healthcare institution, which begins with a spear phishing attack that introduces malware into medical devices. By exploiting the vulnerabilities of these devices, the attackers form an APT on the network. These attackers wish to steal sensitive data related to patients and disrupt health services. Fig. 13 provides an overview about the multistage cyber-attack scenario discussed below.

B. STEP-BY-STEP INTERACTION OF SOC COMPONENTS

1) INITIAL PHISHING ATTACK DETECTION AND RESPONSE

- **Event:** Email scams with malicious attachments, which some employees open.
- **SIEM System:** The SIEM system sources email gateways, endpoint security logs, and network traffic. AI-powered NLP models analyze the email content to identify phishing attempts. The SIEM system thus recognizes the sudden increase in suspicious email activities and highlights the workstation that opened the malicious attachment. An alert will show possible malware infiltration during the process.

- **Intrusion Detection and Response:** In real-time, this IDR system checks the infected workstation for its peculiarity of activities. AI models analyze the process execution, network connections, and file changes of workstations. The IDR system detects malware that connects to an external command and controls the C2 server. Then, it isolates the workstation and blocks outbound connections, preventing further malware communication.
- **RL Layer:** The RL layer automates the response to isolate the workstation and scan other systems. The RL layer refines its response strategy based on the effectiveness of containment in improving future responses to similar incidents.

2) MALWARE SPREAD AND CONTAINMENT

- **Event:** After being contained for a given time, the malware propagates to other devices via vulnerable medical equipment that is outdated, connected, and running obsolete software.
- **IDM:** The IDM module detects malware spread and mitigates its proliferation. Predictive ML models evaluate the probable infection paths using network topology and knowledge of device vulnerabilities. Compromised medical devices are identified by the IDM system, followed by segmentation from the leading network, rerouting, and traffic patching to close vulnerabilities.
- **SIEM System:** The SIEM system correlates data from medical devices, network logs, and IDM system. It updates its threat landscape, integrating the newly discovered vulnerabilities and infection paths into its monitoring algorithms.

3) THREAT INTELLIGENCE AND APT DETECTION

- **Event:** On realizing that the malware has already been contained, the attacker will be prompted to launch an APT to ensure long-term access to the network.
- **Threat Intelligence:** The internal and external threat data are analyzed to identify IoCs associated with known APT groups. AI-driven analysis correlates threat data worldwide with network activity in the institution. In this case, the threat intelligence component identifies that the tactics used in the attack are the same as those from a known APT group that targeted healthcare institutions. It then shares these data with the SIEM and APT detection systems.
- **APT Detection:** The APT detector monitors subtle, long-term attack patterns that are characteristics of APTs. The system uses deep learning models to analyze indicators such as low-frequency network traffic and atypical lateral movement within the network. APT detectors detect hidden backdoors and unauthorized access attempts that bypass all other security solutions while relating that activity to threat intelligence data feeds, thus verifying the APT.

- **IDR:** IDR fires back in response to the detected APT, thus isolating the affected segments and throwing them into a forensic experiment. The RL layer automates the isolation and remediation process; therefore, it rapidly and effectively responds to the attack.

4) PROACTIVE DEFENSE VIA INTRUSION PREDICTION

- **Event:** Although the first set of threats has been minimized, potential attacks continuously threaten the institution.
- **Intrusion Prediction:** This component analyses trends and patterns to predict the possibility of future potential threats. The ML models analyze the attack data history, network behavior, and threat intelligence to predict attack vectors. The intrusion prediction system predicts a possible attack that can be performed against the electronic health record systems in hospitals based on the trends from recent threats and the current threat landscape.
- **SIEM System:** The SIEM system realigns network readiness by fine-tuning monitoring thresholds and heightening the monitoring level of suspicious activities about the forecast attack vector. This increases the readiness of the network for threat detection and response in cases wherein the forecast attack occurs, ensuring that the actions are taken in time.
- **IDM:** The IDM system implements pre-empted mitigating measures such as securing backup systems, tightening access control, and monitoring critical assets. These measures lower the vulnerability of the institution to the predicted attack and boosts its resilience.

5) CONTINUOUS IMPROVEMENT WITH THE RL LAYER

- **RL Layer:** This layer provides constant feedback to all SOC components and optimizes their operation in addition to automating some repetitive tasks. AI and ML models learn from previous incidents and adapt detection thresholds, response times, and mitigation strategies. The RL layer reduces the manual workload of SOC analysts by automatically detecting routines and responses. It simultaneously adjusts the SOC operations based on the latest threat data and experiences.
- **SOC Component Interaction:** SIEM, IDR, IDM, threat intelligence, APT detection, and intrusion prediction systems continuously exchange data and insights to create a unified adaptive defense system that evolves with new and emerging threats. Such robust cooperation ensures that a healthcare provider is immune to various attacks, starting from the first intrusion attempts and ending with complex, long-standing ones.

C. USE CASE STUDY CONCLUSION

This use case describes how each SOC component contributes to the security of the institution.



FIGURE 14. Challenges in the SOC.

- **SIEM System:** This is the central component in an SOC that can correlate data to provide visibility to the entire network. It can thus draw on all the other SOC components to prioritize the effective management of security alerts.
- **IDR and IDM Systems:** These layers detect and contain immediate threats, and the threat intelligence and APT detection components add depth to analyze and respond to sophisticated, long-term attacks.
- **Intrusion Prediction:** This component allows for a proactive stance, thus making the institution ready to handle any future threats by predicting the vectors through which an attack might occur. It also enables taking possible remedial, preemptive steps toward preventing it.
- **RL Layer:** This layer links all these components and optimizes them within a learning framework, in which automated responses learn from past incidents and adapt to the evolution of threats.

These components together develop into a full-scale security framework to protect patient data against leakage or theft,

continue the healthcare cycle, and maintain compliance with any regulatory standard. The integrated SOC is an important, resilient, proactive security posture that protects the healthcare institution from multifaceted cyber-attacks.

VIII. CHALLENGES AND FUTURE TRENDS IN SECURITY OPERATIONS CENTERS

In this section, we investigate **RQ4** by exploring the current challenges faced by SOC and examining future trends. This analysis provides insights into how these challenges can impact overall security effectiveness and how organizations can adapt to evolving threats and technological advancements.

A. CHALLENGES IN SOCs

Managing an SOC is a cumbersome task, as several responsibilities must be balanced to ensure an organization remains safe from cyber threats. Although SOCs are the front line of defense, they have several drawbacks of varying dimensions and nature. The challenges of a SOC, from data management to real-time threat detection, should be tackled in advance so

that it can work optimally. Fig. 14 provides a high level look at the different challenges in SOC. Some critical challenges surrounding the SOC are discussed below:

1) DATA MANAGEMENT

- **Data Overload:** Modern-day SOC's are massively deluged with data. The ongoing security events pose significant difficulty to SOC's in keeping up with relevant events. Liu et al. [123] mentioned that SOC's need better ways to deal with this data deluge.
- **Data Completeness and Consistency:** Incomplete and inconsistent SOC data resemble missing or mismatched pieces of a jigsaw puzzle. Chhetri et al. [126] highlighted that the data must be clean and reliable to ensure good security decisions.
- **Data Validation Mechanisms:** Data validation involve quality check of data before use. For instance, [4] have stressed the need for automatic checks for appropriateness to ascertain that the data that SOC's rely heavily upon are trustworthy.
- **Data Quality Control Mechanisms:** Data quality is essential to avoid misfires and detect potential threats. Ofte and Katsikas [16] reported that an SOC may find it challenging to maintain appropriate checks on threats without effective quality control.

Data management in SOC presents significant challenges that can compromise its effectiveness in responding to cyber threats. One of the primary issues is data overload, where SOC's are overwhelmed by vast amounts of information from various security events, making it difficult for analysts to search for and identify relevant incidents. This overwhelming volume can lead to overlooked critical alerts. Additionally, the completeness and consistency of data play a crucial role in effective decision-making. Incomplete or inconsistent data can resemble a puzzle with missing pieces, thus preventing the ability to form a clear picture of the security landscape. To address these issues, it is essential to implement robust data-validation mechanisms. These mechanisms ensure that data are subjected to quality checks before use, thereby enhancing its trustworthiness. Furthermore, effective data quality control measures are necessary to avoid errors in threat detection. Without proper checks, SOC's may struggle to accurately assess threats, leading to potential vulnerabilities.

2) ALERT MANAGEMENT

- **Alert Fatigue:** Similar to cases wherein excessive notifications are ignored, SOC analysts experience alert fatigue. Wang et al. [13] discussed using AI to help reduce this overwhelming noise.
- **False Positives:** A repetitive false alarm can be frustrating. Huang and Zhu [129] are working on more intelligent systems to reduce these unnecessary alerts.
- **Alert Prioritization:** As all alerts are not equal, prioritizing the ones to be addressed first remains challenging.

Chavali et al. [32] proposed methods to prioritize such alerts to ensure that most critical ones immediately attract attention.

Alert fatigue is a significant concern in SOC, where analysts are often overwhelmed by excessive notifications, leading to important alerts being ignored. This phenomenon occurs when the volume of alerts exceeds analysts' capacity to respond effectively, ultimately diminishing their ability to detect genuine threats. To counter this issue, artificial intelligence has been proposed as a solution to filter out noise and reduce the burden on the analysts. Additionally, false positives further exacerbate alert fatigue because repetitive false alarms can lead to frustration and desensitization among SOC security analysts. The development of more intelligent systems aimed at minimizing unnecessary alerts is essential for maintaining focus on genuine threats. Furthermore, alert prioritization remains a challenge for SOC's because not all alerts carry the same level of urgency. Implementing effective prioritization methods is crucial to ensure that the most critical alerts receive immediate attention, allowing analysts to concentrate their efforts where they are most needed. Addressing the issues of alert fatigue, false positives, and prioritization, can enhance SOC operational efficiency and improve response to security incidents.

3) INCIDENT RESPONSE

- **Incident Response Time:** Responding to security incidents quickly is essential. The faster the response, the less damage is precluded. Naseer et al. [130] explained how real-time analytics can be critical in helping SOC's respond quickly.
- **Automating Responses:** Given the vast number of incidents, automating responses against routine ones is essential to save time for more complex issues. [4] proposed maximum automation to facilitate smooth functioning of SOC's.
- **Mitigation Accuracy:** Accuracy is paramount when handling threats. Tsakalidis et al. [125] discussed on improving the accuracy of SOC's in threat identification and neutralization.

The incident response time is a critical factor in effective cybersecurity management because a swift response to security incidents can significantly reduce potential damage. The quicker an SOC can react, the better the chance of mitigating the impact of an attack. Real-time analytics plays a vital role in enabling SOC's to respond promptly, allowing them to identify and address threats as they arise. Automating responses to routine incidents is essential for optimizing operational efficiency. With the high volume of incidents that SOC's encounter, automation can help manage these repetitive tasks, freeing analysts to focus on more complex and nuanced issues that require human intervention. Furthermore, mitigation accuracy is paramount in threat management. Ensuring the precise identification and neutralization of threats is crucial for maintaining security integrity. Improving the accuracy

of detection and response strategies within SOC helps to minimize the risk of overlooking genuine threats or misidentifying benign activities as malicious. Emphasizing rapid incident responses, automation of routine tasks, and enhanced accuracy in threat handling, can significantly enhance effectiveness in safeguarding against cyber threats.

4) INTEGRATION

- **AI and ML integration:** AI and ML will revolutionize SOC in threat detection, thus enabling effective responses to threats. Nguyen et al. [107] reported that such technologies are shifting the nature of SOC.
- **Security Tools Integration:** SOC depend on various tools, which must be integrated for smooth functioning. Rosso et al. [128] reported the best practices used for integrating such tools within SOC.
- **Interoperability:** For SOC to be effective, disparate systems must communicate fluently with one another without issues. Tsakalidis et al. [125] emphasized the need to ensure that all the components of a SOC work together harmoniously.

The integration of AI and ML is poised to transform the SOC by enhancing threat detection capabilities and enabling more effective responses to cyber threats. These advanced technologies fundamentally shift the operational landscape of SOC, allowing for the faster and more accurate identification of potential security incidents. In addition, SOC rely on a variety of security tools that must be seamlessly integrated to ensure smooth and efficient operations. Best practices for tool integration are essential for streamlining processes and improving overall effectiveness. Furthermore, the interoperability among different systems is crucial for the success of SOC. For SOC to operate efficiently, it is vital that disparate systems communicate effectively without technical barriers. Ensuring that all components of an SOC work together harmoniously enhance collaboration and improves the ability to respond to threats. Focusing on AI and ML integration, effective security tool integration, and ensuring interoperability, can boost SOC operational capabilities and resilience to evolving cyber threats.

5) THREAT INTELLIGENCE

- **Threat Intelligence Utilization:** The more the SOC leverages threat intelligence, the more it stays ahead of adversaries. Haass [1] explained the practical application of this approach to enhance security.
- **Threat Data Correlation:** Data collected from various sources provide a more precise context of threats. Islam et al. [138] presented the criticality of data correlation for accurate threat detection.
- **Integration of External Threat Feeds:** The consumption of external threat intelligence adds an extra level of security. Chhetri et al. [126] elaborated on how SOC can incorporate such helpful intelligence.

Utilizing threat intelligence is vital for the SOC to maintain a proactive stance against adversaries. By effectively leveraging threat intelligence, SOC can stay ahead of potential threats and enhance their overall security. Practical applications of this approach can significantly improve the decision-making process for identifying and mitigating risks. Additionally, correlating threat data collected from various sources is crucial for establishing a more precise context for potential threats. The ability to synthesize and analyze diverse data points allows SOC to achieve accurate threat detection, as highlighted by research on the importance of data correlation. Moreover, integrating external threat feeds into SOC operations adds an additional security layer. By incorporating external threat intelligence, SOC can access valuable insights that enhance their understanding of threat landscapes. This integration allows for a more comprehensive view of the potential risks and enables SOC to respond more effectively. By focusing on threat intelligence utilization, data correlation, and integration of external threat feeds, SOC can significantly bolster their capabilities in identifying and addressing cybersecurity threats.

6) APT DETECTION

- **APT Detection:** APTs are the deadliest and silent of all attacks; therefore, their detection is complicated. Naseer et al. [130] proposed new methods for the early detection of these threats.
- **Long-Term Attack Monitoring:** Some threats do not appear the way they are immediately and have to be tracked over time. Ofte and Katsikas [16] discussed the necessity for constant monitoring to identify such slow-stepping threats.
- **APT Mitigation Strategies:** After an APT is detected, it must be mitigated. Rosso et al. [128] proposed new strategies for mitigating such a highly persistent threat.

Detecting APT poses a significant challenge for SOC because of its stealthy and deadly nature. The complexity of APT detection necessitates the development of innovative methods for identifying these threats. Recent studies have proposed new techniques aimed at enhancing the ability to detect APTs before they can cause substantial harm. Additionally, long-term attack monitoring is essential because some threats may not manifest immediately and require ongoing surveillance to identify their presence. Continuous monitoring is crucial for recognizing slow-stepping threats, as highlighted by discussions of the importance of maintaining vigilance over time. Furthermore, once an APT is detected, effective mitigation strategies must be employed to neutralize the threat. The development of new approaches to mitigate such highly persistent threats is vital to ensure that SOC can respond appropriately and effectively. Focusing on APT detection, the need for long-term monitoring, and the implementation of robust mitigation strategies, can enhance SOC defenses against sophisticated and enduring cyber threats.

7) COMPLIANCE

- **Regulatory Compliance:** SOC keep up with current regulations. Feher and Nguyen et al. [107] discussed the compliance of SOC with changing laws and the need for such a mechanism.
- **Legal Challenges:** Dealing with legal challenges is another major challenge, particularly in data privacy. Spring and Illari [14] highlighted many legal challenges that SOC must face.
- **Sensitive Data Handling:** Protecting sensitive information is the topmost priority of SOC. Tsakalidis et al. [125] discussed data security and privacy in these environments.

Regulatory compliance is a critical aspect of SOC, as it must stay updated with the current laws and regulations. The ability to adapt to changing legal requirements is essential to maintaining operational integrity and avoiding penalties. In addition, SOC face significant legal challenges, particularly concerning data privacy. The handling of sensitive data remains a top priority of SOC. Protecting confidential information is crucial not only for compliance but also for maintaining trust with clients and stakeholders. Addressing regulatory compliance, addressing legal challenges, and ensuring secure handling of sensitive data, will have big impact on SOC effectiveness and resilience in the face of evolving legal and regulatory landscapes.

8) WORKFORCE

- **Skilled Workforce Shortage:** There is a shortage of cybersecurity experts. Chhetri et al. [126] delved into how AI can help fill this gap by automating some tasks.
- **Continuous Training and Development:** SOC personnel must be trained and updated with the latest trends. Liu et al. [123] argued that ongoing training programs must run harmoniously with AI trends.
- **Human Intervention vs. Automation:** For efficient functioning of SOC, human decision-making and automation must be balanced. Chhetri et al. [126] proposed frameworks for effective human-AI collaboration.

The shortage of skilled workforce in cybersecurity poses a significant challenge in SOC. With a limited number of available experts, organizations are increasingly turning to AI to help bridge this gap by automating routine tasks and improving operational efficiency. Continuous training and development are essential for SOC security analysts to stay up-to-date with the latest trends and technologies in the field. Implementing ongoing training programs ensures that staff members are well equipped to handle emerging threats and utilize new tools effectively. Additionally, finding the right balance between human intervention and automation is crucial for efficient functioning of SOC. Although automation can streamline processes and enhance productivity, human decision-making remains vital for navigating complex security scenarios. Developing frameworks for effective

collaboration between human analysts and AI systems can help SOC leverage the strengths of both, ensuring a robust and responsive security posture. By addressing the shortage of skilled workforce, prioritizing continuous training, and fostering human-AI collaboration, SOC can significantly enhance their capabilities to defend against cyber threats.

9) MONITORING

- **Continuous Monitoring:** SOC should engage in continuous monitoring to detect real-time threats. Naseer et al. [130] established the importance of constant monitoring systems in this regard.
- **24/7 Coverage:** Continuous monitoring of threats is a significant challenge. Liu et al. [123] discussed the use of AI for 24/7 vigilance of such threats.
- **Real-Time Threat Detection:** Real-time detection of threats time is crucial for preventing damage. Huang and Zhu [129] explored on increasing the detection coverage in real-time SOC.

Continuous monitoring is essential for the SOC to effectively detect real-time threats and promptly respond to potential security incidents. Engaging in constant monitoring systems allows SOC to maintain vigilance over their networks and to identify anomalies as they occur. However, achieving 24/7 coverage presents a significant challenge for many organizations. To address this, the integration of AI can enhance the ability of SOC to maintain round-the-clock vigilance, ensuring that threats are continuously monitored without gaps in coverage. Furthermore, real-time threat detection is crucial for preventing damage from cyberattacks. The ability to identify threats as they emerge enables SOC to take swift action and minimize potential harm. Increasing the detection coverage in real time is vital for enhancing the overall security posture of SOC. Prioritizing continuous monitoring, leveraging AI for 24/7 coverage, and focusing on real-time threat detection, will improve SOC effectiveness in safeguarding themselves against cyber threats.

10) SCALABILITY

- **Scalability of SOC Operations:** SOC operational activities must correspond with the increasing size of networks. Tsakalidis et al. [125] outlined how SOC can be scaled up to handle more extensive and complex environments.
- **Adapting to Network Growth:** SOC have to evolve with increasing network sizes. Rosso et al. [128] provided strategies for expanding SOC capabilities in line with network growth.
- **Extension of the Monitoring Capabilities:** As threats continue to increase, the SOC have to extend their monitoring circles. Liu et al. [123] suggested increasing monitoring without reducing effectiveness.

The scalability of SOC operations is a crucial factor, as organizations expand, and their networks grow larger and more complex. SOC must ensure that their operational activities

align with the increasing size of networks, allowing them to manage and secure a broader range of assets effectively. Adapting to network growth is essential for SOC to remain relevant and effective in their security. Implementing strategies to expand the SOC capabilities in tandem with network growth enables them to address emerging challenges and maintain robust security measures. Additionally, as cyber threats continue to proliferate, SOC must extend their monitoring capabilities to cover a wider array of potential vulnerabilities. This extension of monitoring is vital for ensuring the comprehensive oversight of the network environment. However, it is equally important to increase monitoring capabilities without compromising effectiveness and maintaining a high level of performance, as the scope expands. Focusing on scalability, adapting to network growth, and enhancing monitoring capabilities, will safeguard increasingly complex network infrastructure.

11) COST MANAGEMENT

- **Budget Constraints:** SOC must function with minimal budget. Chhetri et al. [126] discussed the cost-efficient running of SOC.
- **Operations on a Budget:** The biggest challenge with SOC is their effective operation while avoiding overspending. Islam et al. [118] provided frameworks for reducing the associated costs while maintaining security.
- **Resource Allocation:** SOC must use limited resources optimally. Rosso et al. [128] suggested methods for prioritizing resource allocation.

Budget constraints are a significant challenge for SOC, as they require them to operate effectively with minimal financial resources. To remain efficient while managing costs, SOC must implement cost-effective strategies that maximize their operational capabilities without compromising security. Operating within a budget presents the challenge of ensuring that security measures are effective while avoiding unnecessary expenditure. Developing frameworks to reduce the associated costs while maintaining robust security is essential for sustainable SOC operations. Additionally, optimal resource allocation is critical for SOC, because they must make most of their limited resources. Prioritizing resource allocation helps ensure that key areas receive the necessary attention and funding, thus enabling SOC to operate more efficiently. Dealing with budget constraints, focusing on operations within budgetary limits, and optimizing resource allocation, can enhance SOC overall effectiveness and resilience in the face of financial challenges.

B. FUTURE TRENDS IN SOC

In the future, SOC will be guided by various trends for addressing the aforementioned challenges and keep up with the evolving landscape of cybersecurity.

1) INCREASED AUTOMATION AND ORCHESTRATION

Automation and orchestration within SOC will be the focus in future to combat alert fatigue and shortage of skilled per-

sonnel. For instance, AI, ML, and robotic process automation will help SOC automate mundane tasks such as alert triage and incident response. This will free up analysts to handle more complex and strategic activities. Automation will, in the future, standardize workflows and playbooks that will help SOC teams respond more efficiently and quicker in cases of potential threats.

2) ADOPTION OF EXTENDED DETECTION AND RESPONSE (XDR)

Extended detection and response, or XDR, is one of the main trends of SOC. In XDR, approaches are integrated with threat detection and response methods. XDR solutions combine data from all parts of the security stack endpoints, networks, and cloud environments onto a single platform. This will assist SOC analysts to view threats holistically and enable effective complex attack detection across domains, allowing for more context during incident response.

3) ADVANCEMENTS IN THREAT INTELLIGENCE INTEGRATION

In future, SOC will witness the enhancement of the implementation of threat intelligence solutions in operation. AI and ML will remain a core component of improving and correlating threat intelligence from various sources such that SOC can better assess and prepare for new threats. The constant monitoring of the activities of adversaries will help SOC foresee attackers' intentions using big data real-time intelligence.

4) EMPHASIS ON PROACTIVE DEFENSE AND THREAT HUNTING

SOC will shift toward a more proactive defense model, and threat hunting will become more fundamental to this function. Analysts can perform threat hunting, i.e., they look for threats even in cases wherein an issue has not been identified. This proactive approach will be accompanied by AI- and ML-based tools that help to discover patterns and outliers characterizing adversarial actions.

5) ENHANCED FOCUS ON DATA PRIVACY AND COMPLIANCE

With increasing complexity of regulatory environments, SOC will be forced to direct their efforts toward data privacy and compliance. Future SOC will employ new generations of encryption, anonymization methodologies, and privacy-preserving analytics to ensure their operations comply with SOX and international regulations. Moreover, constant monitoring and auditing will be the key strategies in long-term compliance and preventing data breach incidents.

6) EVOLUTION OF SOC-AS-A-SERVICE

SOCaaS will gain popularity as organizations seek more flexible and affordable solutions to address cybersecurity issues. SOCaaS solutions are implemented using managed detection and response services, through which organizations

can outsource their SOC activities to specialist teams. This model can be used by small and medium enterprises, which may not have specialized personnel to design and sustain a traditional SOC.

IX. CONCLUSION

One of the biggest challenges facing Security Operation Centers (SOCs) is dealing with the increasingly sophisticated threats. As the threats landscape expand, it become imperative for SOCs to have integrated complex security solution. To address this, we conducted a systematic literature review to identify the current technologies used by different SOC tools. We discussed the proposed approach and limitations of each study. Additionally, we proposed a reference comprehensive architecture for Next Generation SOCs by integrating state-of-the-art AI and ML techniques into various security components. Our proposed architecture deals with interdependent issues in data management, incident response, and threat intelligence, providing enterprises with a method for adequate security against current and future threats. The use case described in this paper demonstrates the value of a holistic approach wherein all the SOC components work together to identify, block, and mitigate cyberattacks. SOCs use AI and ML to maintain a highly proactive security stance to respond to present attacks and to foresee future threats. AI and ML are instrumental tools and technologies employed for the development of SOCs. Future SOCs must be flexible and adaptable to accommodate all these state-of-the-art technologies and methodologies. However, SOC personnel will need to work with the complexity of modern cybersecurity. Moreover, cybersecurity analysts and AI systems will have to work harmoniously to balance everything. In other words, a well-organized and flexible SOC cannot be overestimated, as the need for businesses to protect their digital assets increases in challenging environments.

REFERENCES

- [1] J. C. Haass, "Cyber threat intelligence and machine learning," *IEEE Xplore*, vol. 2, no. 1, pp. 156–159, Sep. 2022.
- [2] N. Gupta, I. Traoré, and P. M. F. D. Quinan, "Automated event prioritization for security operation center using deep learning," *IEEE Xplore*, vol. 2, no. 1, pp. 5864–5872, Dec. 2019.
- [3] S. Rass, S. König, J. Wachter, V. Mayoral-Vilches, and E. Panaousis, "Game-theoretic APT defense: An experimental study on robotics," *Comput. Secur.*, vol. 132, Sep. 2023, Art. no. 103328.
- [4] Z. T. Sworna, C. Islam, and M. A. Babar, "APIRO: A framework for automated security tools API recommendation," *ACM Trans. Softw. Eng. Methodol.*, vol. 32, no. 1, pp. 1–42, Jan. 2023.
- [5] J. Kinyua and L. Awuah, "AI/ML in security orchestration, automation and response: Future research directions," *Intell. Autom. Soft Comput.*, vol. 28, no. 2, pp. 527–545, 2021.
- [6] C. H. Chi, S. Y. Ooi, E. H. Binti, Y. H. Pang, M. K. B. A. Yan, and K. I. B. Sidin, "Intelligent-based SIEM security email alert," in *Proc. 11th Int. Conf. Inf. Commun. Technol. (ICICT)*, Aug. 2023, pp. 481–486.
- [7] H. V. Vo, H. P. Du, and H. N. Nguyen, "AI-powered intrusion detection in large-scale traffic networks based on flow sensing strategy and parallel deep analysis," *J. Netw. Comput. Appl.*, vol. 220, Nov. 2023, Art. no. 103735.
- [8] G. González-Granadillo, M. Faiella, I. Medeiros, R. Azevedo, and S. González-Zarzosa, "ETIP: An enriched threat intelligence platform for improving OSINT correlation, analysis, visualization and sharing capabilities," *J. Inf. Secur. Appl.*, vol. 58, May 2021, Art. no. 102715.
- [9] A. Sridharan and V. Kanchana, "SIEM integration with SOAR," in *Proc. Int. Conf. Futuristic Technol. (INCOFT)*, Nov. 2022, pp. 1–6.
- [10] K. Hughes, K. McLaughlin, and S. Sezer, "A model-free approach to intrusion response systems," *J. Inf. Secur. Appl.*, vol. 66, May 2022, Art. no. 103150.
- [11] T. Ahmed, A. Shah, M. Kolla, and R. Yellasi, "Reduction of alert fatigue using extended isolation forest," in *Proc. Int. Conf. Forensics, Analytics, Big Data, Secur. (FABS)*, Dec. 2021, pp. 1–5.
- [12] K. Prabu and P. Sudhakar, "An automated intrusion detection and prevention model for enhanced network security and threat assessment," *Int. J. Comput. Netw. Appl.*, vol. 10, no. 4, p. 621, Aug. 2023.
- [13] X. Wang, X. Yang, X. Liang, X. Zhang, W. Zhang, and X. Gong, "Combating alert fatigue with AlertPro: Context-aware alert prioritization using reinforcement learning for multi-step attack detection," *Comput. Secur.*, vol. 137, Feb. 2024, Art. no. 103583.
- [14] J. M. Spring and P. Illari, "Review of human decision-making during computer security incident analysis," *Digit. Threats, Res. Pract.*, vol. 2, no. 2, pp. 1–47, Jun. 2021.
- [15] J. R. Goodall, E. D. Ragan, C. A. Steed, J. W. Reed, G. D. Richardson, K. M. T. Huffer, R. A. Bridges, and J. A. Laska, "Situ: Identifying and explaining suspicious behavior in networks," *IEEE Trans. Vis. Comput. Graphics*, vol. 25, no. 1, pp. 204–214, Jan. 2019.
- [16] H. J. Ofte and S. Katsikas, "Understanding situation awareness in SOCs, a systematic literature review," *Comput. Secur.*, vol. 126, Mar. 2023, Art. no. 103069.
- [17] O. Akinrolabu, I. Agrafiotis, and A. Erola, "The challenge of detecting sophisticated attacks," in *Proc. 13th Int. Conf. Availability, Rel. Secur.*, Aug. 2018, pp. 1–9.
- [18] H. Bennouri, A. Abdi, I. Hossain, and A. Pujol, "The role of SOC in ensuring the security of IoT devices: A review of current challenges and future directions," in *Proc. 12th Medit. Conf. Embedded Comput. (MECO)*, Jun. 2023, pp. 1–8.
- [19] Z. T. Sworna, Z. Mousavi, and M. A. Babar, "NLP methods in host-based intrusion detection systems: A systematic review and future directions," *J. Neww. Comput. Appl.*, vol. 220, Nov. 2023, Art. no. 103761.
- [20] S. Neupane, J. Ables, W. Anderson, S. Mittal, S. Rahimi, I. Banicescu, and M. Seale, "Explainable intrusion detection systems (X-IDS): A survey of current methods, challenges, and opportunities," *IEEE Access*, vol. 10, pp. 112392–112415, 2022.
- [21] Y. T. Dun, "Grasp on next-generation security operation center (NGSOC): Comparative study," *Int. J. Nonlinear Anal. Appl.*, vol. 12, pp. 2008–6822, May 2021.
- [22] N. Afzaliseresht, Y. Miao, S. Michalska, Q. Liu, and H. Wang, "From logs to stories: human-centred data mining for cyber threat intelligence," *IEEE Access*, vol. 8, pp. 19089–19099, 2020.
- [23] M. Saraiva and N. Mateus-Coelho, "CyberSoc framework a systematic review of the state-of-art," *Proc. Comput. Sci.*, vol. 204, pp. 961–972, Apr. 2022.
- [24] D. Moher, A. Liberati, J. Tetzlaff, and D. G. Altman, "Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement," *Int. J. Surg.*, vol. 339, no. 1, p. 2535, Jul. 2009.
- [25] T. Yang, Y. Qiao, and B. Lee, "Towards trustworthy cybersecurity operations using Bayesian deep learning to improve uncertainty quantification of anomaly detection," *Comput. Secur.*, vol. 144, Sep. 2024, Art. no. 103909.
- [26] T. Kwon, J.-s. Song, S. Choi, Y. Lee, and J. Park, "VISNU: A novel visualization methodology of security events optimized for a centralized SOC," in *Proc. 13th Asia Joint Conf. Inf. Secur. (AsiaJCIS)*, Aug. 2018, pp. 1–7.
- [27] B. Song, J. Choi, S.-S. Choi, and J. Song, "Visualization of security event logs across multiple networks and its application to a CSOC," *Cluster Comput.*, vol. 22, no. S1, pp. 1861–1872, Jan. 2019.
- [28] W. Wang, P. Yi, J. Jiang, P. Zhang, and X. Chen, "Transformer-based framework for alert aggregation and attack prediction in a multi-stage attack," *Comput. Secur.*, vol. 136, Jan. 2024, Art. no. 103533.
- [29] Ö. Sen, D. van der Velde, K. A. Wehrmeister, I. Hacker, M. Henze, and M. Andres, "On using contextual correlation to detect multi-stage cyber attacks in smart grids," *Sustain. Energy, Grids Netw.*, vol. 32, Dec. 2022, Art. no. 100821.
- [30] R. Vaarandi and A. Guerra-Manzanaraes, "Stream clustering guided supervised learning for classifying NIDS alerts," *Future Gener. Comput. Syst.*, vol. 155, pp. 231–244, Jun. 2024.

- [31] B. Munsinger, N. Beebe, and T. Richardson, "Virtual reality for improving cyber situational awareness in security operations centers," *Comput. Secur.*, vol. 132, Sep. 2023, Art. no. 103368.
- [32] L. Chavali, A. Krishnan, P. Saxena, B. Mitra, and A. Sreevallabh Chivukula, "Off-policy actor-critic deep reinforcement learning methods for alert prioritization in intrusion detection systems," *Comput. Secur.*, vol. 142, Jul. 2024, Art. no. 103854.
- [33] J. H. Ring, C. M. Van Oort, S. Durst, V. White, J. P. Near, and C. Skalka, "Methods for host-based intrusion detection with deep learning," *Digit. Threats, Res. Pract.*, vol. 2, no. 4, pp. 1–29, Dec. 2021.
- [34] D.-M. Ngo, D. Lightbody, A. Temko, C. Pham-Quoc, N.-T. Tran, C. C. Murphy, and E. Popovici, "HH-NIDS: Heterogeneous hardware-based network intrusion detection framework for IoT security," *Future Internet*, vol. 15, no. 1, p. 9, Dec. 2022.
- [35] S. A. Alharbi. (2021). *A Qualitative Study on Security Operations Centers in Saudi Arabia: Challenges and Research Directions*.
- [36] R. Vaarandi and A. Guerra-Manzanares, "Network IDS alert classification with active learning techniques," *J. Inf. Secur. Appl.*, vol. 81, Mar. 2024, Art. no. 103687.
- [37] T. Mézesová, P. Sokol, and T. Bajtos, "Evaluation of attackers' skill levels in multi-stage attacks," *Information*, vol. 11, no. 11, p. 537, Nov. 2020.
- [38] J. Happa, I. Agraftiotis, M. Helmhout, T. Bashford-Rogers, M. Goldsmith, and S. Creese, "Assessing a decision support tool for SOC analysts," *Digit. Threats, Res. Pract.*, vol. 2, no. 3, pp. 1–35, Sep. 2021.
- [39] R. van der Kleij, J. M. Schraagen, B. Cadet, and H. Young, "Developing decision support for cybersecurity threat and incident managers," *Comput. Secur.*, vol. 113, Feb. 2022, Art. no. 102535.
- [40] I. Wawrowski, M. Michalak, A. Białas, R. Kurianowicz, M. Sikora, M. Uchrowski, and A. Kajzer, "Detecting anomalies and attacks in network traffic monitoring with classification methods and XAI-based explainability," *Proc. Comput. Sci.*, vol. 192, pp. 2259–2268, Apr. 2021.
- [41] C. Engel, S. Mencke, R. Heumuller, R. Hormann, H. Aedtner, and F. Ortmeier, "Customizable operation center for smart security management," *Proc. CIRP*, vol. 104, pp. 1930–1935, Apr. 2021.
- [42] Y. Kanemoto, K. Aoki, M. Iwamura, J. Miyoshi, D. Kotani, H. Takakura, and Y. Okabe, "Detecting successful attacks from IDS alerts based on emulation of remote shellcodes," in *Proc. IEEE 43rd Annu. Comput. Softw. Appl. Conf. (COMPSAC)*, Jul. 2019, pp. 471–476.
- [43] S. Latif, W. Boulila, A. Koubaa, Z. Zou, and J. Ahmad, "DTL-IDS: An optimized intrusion detection framework using deep transfer learning and genetic algorithm," *J. Netw. Comput. Appl.*, vol. 221, Jan. 2024, Art. no. 103784.
- [44] P. Jaroensiripong, K. Sumongkayothin, P. Siritanawan, and K. Kotani, "Cybersecurity intrusion detection with image classification model using Hilbert curve," in *Proc. 19th Int. Joint Conf. Comput. Vis., Imag. Comput. Graph. Theory Appl.*, 2024, pp. 325–332.
- [45] A. Presekal, A. Stefanov, V. S. Rajkumar, and P. Palensky, "Cyber forensic analysis for operational technology using graph-based deep learning," in *Proc. IEEE Int. Conf. Commun., Control, Comput. Technol. Smart Grids*, 2023, pp. 1–15.
- [46] M. Wu and Y. Moon, "Alert correlation for cyber-manufacturing intrusion detection," *Proc. Manuf.*, vol. 34, pp. 820–831, May 2019.
- [47] A. Shah, R. Ganesan, S. Jajodia, and H. Cam, "A two-step approach to optimal selection of alerts for investigation in a CSOC," *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 7, pp. 1857–1870, Jul. 2019.
- [48] C. Zhong, A. Alnusair, B. Sayger, A. Troxell, and J. Yao, "AOH-map: A mind mapping system for supporting collaborative cyber security analysis," *IEEE Xplore*, vol. 10030, no. 2, pp. 74–80, Apr. 2019.
- [49] D. Javeed, M. S. Saeed, I. Ahmad, P. Kumar, A. Jolfaei, and M. Tahir, "An intelligent intrusion detection system for smart consumer electronics network," *IEEE Trans. Consum. Electron.*, vol. 6, no. 1, pp. 1–16, Apr. 2023.
- [50] M. Vermeer, N. Kadenko, M. van Eeten, C. Gañán, and S. Parkin, "Alert alchemy: SOC workflows and decisions in the management of NIDS rules," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2023, pp. 2770–2784.
- [51] Y. Jiang and Y. Atif, "A selective ensemble model for cognitive cybersecurity analysis," *J. Netw. Comput. Appl.*, vol. 193, Nov. 2021, Art. no. 103210.
- [52] S. Iglesias Pérez, S. Moral-Rubio, and R. Criado, "A new approach to combine multiplex networks and time series attributes: Building intrusion detection systems (IDS) in cybersecurity," *Chaos, Solitons Fractals*, vol. 150, Sep. 2021, Art. no. 111143.
- [53] R. Kalakoti, S. Namm, and H. Bahsi, "Enhancing IoT botnet attack detection in SOCs with an explainable active learning framework," in *Proc. IEEE World AI IoT Congr. (AIoT)*, May 2024, pp. 265–272.
- [54] A. Shah, R. Ganesan, and S. Jajodia, "A methodology for ensuring fair allocation of CSOC effort for alert investigation," *Int. J. Inf. Secur.*, vol. 18, no. 2, pp. 199–218, Apr. 2019.
- [55] M. Hamad, A. Finkenzeller, M. Kuhr, A. Roberts, O. Maennel, V. Prevelakis, and S. Steinhorst, "REACT: Autonomous intrusion response system for intelligent vehicles," *Comput. Secur.*, vol. 145, Oct. 2024, Art. no. 104008.
- [56] L. Mitchell, "Two can play that game," *ACM Trans. Intell. Syst. Technol.*, vol. 11, no. 4, pp. 36–37, Jan. 2017.
- [57] F. Mocanu and L. Scripcariu, "Implementation of a security operation Center—An essential cybersecurity solution for organizations," in *Proc. 27th Int. Conf. Syst. Theory, Control Comput. (ICSTCC)*, Oct. 2023, pp. 539–544.
- [58] H. G. Shon, Y. Lee, and M. Yoon, "Semi-supervised alert filtering for network security," *Electronics*, vol. 12, no. 23, p. 4755, Nov. 2023.
- [59] E. Agepong, Y. Cherdantseva, P. Reinecke, and P. Burnap, "Towards a framework for measuring the performance of a security operations center analyst," in *Proc. Int. Conf. Cyber Secur. Protection Digit. Services*, Jun. 2020, pp. 1–8.
- [60] H. S. Eriksson and G. Grov, "Towards XAI in the SOC—A user centric study of explainable alerts with SHAP and LIME," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2022, pp. 2595–2600.
- [61] S. Hiruta, S. Ikeda, S. Shima, and H. Takakura, "IDS alert priority determination based on traffic behavior," in *Lecture Notes in Computer Science*, 2019, pp. 189–206.
- [62] J.-Y. Kim and H.-Y. Kwon, "Threat classification model for security information event management focusing on model efficiency," *Comput. Secur.*, vol. 120, Sep. 2022, Art. no. 102789.
- [63] T. D. Bikov, D. Radev, T. Iliev, and D. Stankovski, "Threat hunting as cyber security baseline in the next-generation security operations center," in *Proc. 29th Telecommun. Forum*, Nov. 2021, pp. 1–4.
- [64] L. Axon, J. Happa, A. Janse van Rensburg, M. Goldsmith, and S. Creese, "Sonification to support the monitoring tasks of security operations centres," *IEEE Trans. Dependable Secur. Comput.*, vol. 22, no. 1, pp. 1–20, Apr. 2019.
- [65] B. Arfaoui, H. Mrabet, and A. Jemai, "SOCaaS-IoT" a security operations center as a service approach for IoT applications using open-source SIEM," in *Proc. IEEE Afro-Mediterranean Conf. Artif. Intell.*, Dec. 2023, pp. 1–6.
- [66] A. Hwoij, A. Khamaiseh, and M. Ababneh, "SIEM architecture for the Internet of Things and smart city," in *Proc. Int. Conf. Data Sci., E-Learn. Inf. Syst.*, Apr. 2021, p. 4.
- [67] H. Mokalled, R. Catelli, V. Casola, D. Debertol, E. Meda, and R. Zunino, "The applicability of a SIEM solution: Requirements and evaluation," in *Proc. IEEE 28th Int. Conf. Enabling Technol., Infrastruct. Collaborative Enterprises (WETICE)*, Jun. 2019, pp. 132–137.
- [68] P. Najafi, A. Muhle, W. Punter, F. Cheng, and C. Meinel, "MalRank: A measure of maliciousness in SIEM-based knowledge graphs," in *Proc. 35th Annu. Comput. Secur. Appl. Conf.*, Dec. 2019, pp. 417–429.
- [69] J. C. Sancho, A. Caro, M. Ávila, and A. Bravo, "New approach for threat classification and security risk estimations based on security event management," *Future Gener. Comput. Syst.*, vol. 113, pp. 488–505, Dec. 2020.
- [70] B. D. Bryant and H. Saiedian, "Improving SIEM alert metadata aggregation with a novel kill-chain based classification model," *Comput. Secur.*, vol. 94, Jul. 2020, Art. no. 101817.
- [71] A. Skendžić, B. Kovacic, and B. Balon, "Management and monitoring security events in a Bus. Organization—SIEM system," in *Proc. 45th Jubilee Int. Conv. Inf., Commun. Electron. Technol. (MIPRO)*, May 2022, pp. 1203–1208.

- [72] G. R. McRee, "Improved detection and response via optimized alerts: Usability study," *J. Cybersecurity Privacy*, vol. 2, no. 2, pp. 379–401, May 2022.
- [73] M. Hristov, M. Nenova, G. Iliev, and D. R. Avresky, "Integration of splunk enterprise SIEM for DDoS attack detection in IoT," *IEEE Xplore*, vol. 3, no. 1, pp. 1–5, Nov. 2021.
- [74] M. Dietz, M. Vielberth, and G. Pernul, "Integrating digital twin security simulations in the security operations center," in *Proc. 15th Int. Conf. Availability, Rel. Secur.*, Aug. 2020, pp. 1–15.
- [75] N. Akshai Sankar and K. A. Fasila, "Implementation of SOC using ELK with integration of wazuh and dedicated file integrity monitoring," in *Proc. 9th Int. Conf. Smart Comput. Commun. (ICSCC)*, Aug. 2023, pp. 350–354.
- [76] V. H. Perera, A. N. Senarathne, and L. Rupasinghe, "Intelligent SOC chatbot for security operation center," in *Proc. Int. Conf. Advancements Comput. (ICAC)*, Dec. 2019, pp. 340–345.
- [77] R. Andrade and J. Torres, "Enhancing intelligence SOC with big data tools," in *Proc. IEEE 9th Annu. Inf. Technol., Electron. Mobile Commun. Conf. (IEMCON)*, Nov. 2018, pp. 1076–1080.
- [78] M. Cinque, R. Della Corte, and A. Pecchia, "Contextual filtering and prioritization of computer application logs for security situational awareness," *Future Gener. Comput. Syst.*, vol. 111, pp. 668–680, Oct. 2020.
- [79] R. G. Dimitoglou, "Development of a virtualized security operations center," in *Consortium for Computing Sciences in Colleges*, vol. 37, 2021, pp. 108–119.
- [80] C. Onwubiko, "CoCoa: An ontology for cybersecurity operations center analysis process," *IEEE Xplore*, vol. 1, no. 2, pp. 1–8, Apr. 2018.
- [81] M. S. Khan, "Cyber threat hunting," *Int. J. Cogn. Inform. Nat. Intell.*, vol. 15, pp. 1–23, Apr. 2021.
- [82] A. H. Nursidiq and C. Lim, "Cyber threat hunting to detect unknown threats in the enterprise network," in *Proc. IEEE Int. Conf. Cryptography, Informat., Cybersecurity (ICoCICs)*, Aug. 2023, pp. 303–308.
- [83] S. Ndichu, T. Ban, T. Takahashi, and D. Inoue, "A machine learning approach to detection of critical alerts from imbalanced multi-appliance threat alert logs," in *Proc. IEEE Int. Conf. Big Data*, Dec. 2021, pp. 2119–2127.
- [84] C.-H. Han, "Blockade-detection-response based security operations dashboard design," *Comput. Human Behav. Rep.*, vol. 4, Aug. 2021, Art. no. 100143.
- [85] S. Mihindu and F. Khosrow-shahi, "Collaborative visualization embedded cost-efficient, virtualized cyber security operations center," *IEEE Xplore*, vol. 2, no. 1, pp. 153–159, May 2020.
- [86] M. A. Islam, "Application of artificial intelligence and machine learning in security operations center," School Comput. Fac. Middle Georgia State Univ., Tech. Rep., 2023.
- [87] M. B. M. Hata, M. Y. B. Darus, M. Z. A. B. Shafiee, E. Petrus, and Y. A. Jamian, "A log aggregation design criteria for robust SIEM (security information and event management) in enhancing threat detection," in *Proc. IEEE 8th Int. Conf. Recent Adv. Innov. Eng. (ICRAIE)*, Dec. 2023, pp. 1–6.
- [88] N. Miloslavskaya, "Analysis of SIEM systems and their usage in security operations and security intelligence centers," in *Advances in Intelligent Systems and Computing*, 2018, pp. 282–288.
- [89] W. Tounsi and H. Rais, "A survey on technical threat intelligence in the age of sophisticated cyber attacks," *Comput. Secur.*, vol. 72, pp. 212–233, Jan. 2018.
- [90] A. Calvo, S. Escuder, J. Escrig, M. Arias, N. Ortiz, and J. Guijarro, "A data-driven approach for risk exposure analysis in enterprise security," in *Proc. IEEE 10th Int. Conf. Data Sci. Adv. Analytics (DSAA)*, Oct. 2023, pp. 1–9.
- [91] P. Najafi, F. Cheng, and C. Meinel, "SIEMA: Bringing advanced analytics to legacy security information and event management," in *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, Cham, Switzerland: Springer, 2021, pp. 25–43.
- [92] M. Wurzenberger, G. Höld, M. Landauer, and F. Skopik, "Analysis of statistical properties of variables in log data for advanced anomaly detection in cyber security," *Comput. Secur.*, vol. 137, Feb. 2024, Art. no. 103631.
- [93] K. Demertzis, N. Tziritis, P. Kikiras, S. L. Sanchez, and L. Iliadis, "The next generation cognitive security operations center: Adaptive analytic lambda architecture for efficient defense against adversarial attacks," *Big Data Cognit. Comput.*, vol. 3, no. 1, p. 6, Jan. 2019.
- [94] D. Crooks and L. Velsan. (2019). *Building a Minimum Viable Security Operations Centre for the Modern Grid Environment*.
- [95] A. Ahmad, S. B. Maynard, K. C. Desouza, J. Kotsias, M. T. Whitty, and R. L. Baskerville, "How can organizations develop situation awareness for incident response: A case study of management practice," *Comput. Secur.*, vol. 101, Feb. 2021, Art. no. 102122.
- [96] E. Irshad and A. Basit Siddiqui, "Cyber threat attribution using unstructured reports in cyber threat intelligence," *Egyptian Informat. J.*, vol. 24, no. 1, pp. 43–59, Mar. 2023.
- [97] D. R. Arikkat, V. P., R. R. K. A., S. Nicolazzo, A. Nocera, G. Timpau, and M. Conti, "OSTIS: A novel organization-specific threat intelligence system," *Comput. Secur.*, vol. 145, Oct. 2024, Art. no. 103990.
- [98] D. Skias, S. Tsekeridou, T. Zahariadis, A. Voulkidis, T.-H. Velivassaki, and K. Fotiadou, "Pan-European cybersecurity incidents information sharing platform to support NIS directive," in *Proc. 16th Int. Conf. Availability, Rel. Secur.*, Aug. 2021, pp. 1–7.
- [99] H. Hettema, "Rationality constraints in cyber defense: Incident handling, attribution and cyber threat intelligence," *Comput. Secur.*, vol. 109, Oct. 2021, Art. no. 102396.
- [100] N. Rastogi, S. Dutta, M. J. Zaki, A. Gittens, and C. C. Aggarwal, "MALOnt: An ontology for malware threat intelligence," *Deployable Mach. Learn. Secur. Def.*, vol. 2, pp. 28–44, Jan. 2020.
- [101] W. Yang and K.-Y. Lam, "Automated cyber threat intelligence reports classification for early warning of cyber attacks in next-generation SOC," *Inf. Commun. Secur.*, vol. 1, pp. 145–164, May 2020.
- [102] D. Schlette, M. Vielberth, and G. Pernul, "CTI-SOC2M2—the quest for mature, intelligence-driven security operations and incident response capabilities," *Comput. Secur.*, vol. 111, Dec. 2021, Art. no. 102482.
- [103] F. K. Kaiser, L. J. Andris, T. F. Tennig, J. M. Iser, M. Wiens, and F. Schultmann, "Cyber threat intelligence enabled automated attack incident response," *IEEE Xplore*, vol. 2, no. 1, pp. 1–6, Oct. 2022.
- [104] A. Tundis, S. Ruppert, and M. Muhlhäuser, "A feature-driven method for automating the assessment of OSINT cyber threat sources," *Comput. Secur.*, vol. 113, Feb. 2022, Art. no. 102576.
- [105] R. Muhammad, S. A. Ismail, and N. H. Hassan, "Botnet detection and incident response in security operation center (SOC): A proposed framework," *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 3, p. 389, 2024.
- [106] Y. Wang, Y. Guo, and C. Fang, "An end-to-end method for advanced persistent threats reconstruction in large-scale networks based on alert and log correlation," *J. Inf. Secur. Appl.*, vol. 71, Dec. 2022, Art. no. 103373.
- [107] T. Q. Nguyen, R. Laborde, A. Benzekri, and B. Qu'hen, "Detecting abnormal DNS traffic using unsupervised machine learning," *IEEE Xplore*, vol. 4, no. 3, pp. 1–8, Oct. 2020.
- [108] A. Benzekri, R. Laborde, A. Oglaza, D. Rammal, and F. Barrere, "Dynamic security management drove by situations: An exploratory analysis of logs for the identification of security situations," in *Proc. 3rd Cyber Secur. Netw. Conf.*, 2019, p. 10.
- [109] Y. Ren, Y. Xiao, Y. Zhou, Z. Zhang, and Z. Tian, "CSKG4APT: A cybersecurity knowledge graph for advanced persistent threat organization attribution," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 5695–5709, Jun. 2023.
- [110] L.-Y. Yeh, P. J. Lu, S.-H. Huang, and J.-L. Huang, "SOChain: A privacy-preserving DDoS data exchange service over SOC consortium blockchain," *IEEE Trans. Eng. Manag.*, vol. 67, no. 4, pp. 1487–1500, Nov. 2020.
- [111] P. Alaeifar, S. Pal, Z. Jadidi, M. Hussain, and E. Foo, "Current approaches and future directions for cyber threat intelligence sharing: A survey," *J. Inf. Secur. Appl.*, vol. 83, Jun. 2024, Art. no. 103786.
- [112] S. Anbalagan, G. Raja, S. Gurumoorthy, D. Suresh R, and K. Ayyakannu, "Blockchain assisted hybrid intrusion detection system in autonomous vehicles for Industry 5.0," *IEEE Trans. Consum. Electron.*, vol. 69, no. 4, pp. 881–889, Nov. 2023.
- [113] R. Graf and R. King, "Neural network and blockchain-based technique for cyber threat intelligence and situational awareness," *IEEE Xplore*, vol. 2, no. 1, pp. 409–426, Apr. 2018.
- [114] K. Hughes, K. McLaughlin, and S. Sezer, "Dynamic countermeasure knowledge for intrusion response systems," in *Proc. 31st Irish Signals Syst. Conf. (ISSC)*, Jun. 2020, pp. 1–6.

- [115] A. Coscia, V. Dentamaro, S. Galantucci, A. Maci, and G. Pirlo, "Automatic decision tree-based NIDPS ruleset generation for DoS/DDoS attacks," *J. Inf. Secur. Appl.*, vol. 82, May 2024, Art. no. 103736.
- [116] C. Islam, M. A. Babar, and S. Nepal, "Design and generation of a set of declarative apis for security orchestration," *IEEE Trans. Services Comput.*, vol. 17, no. 1, pp. 127–141, Jan. 2024.
- [117] K. Abeywardena, "Automating threat advisory report generation using threat intelligence," in *Proc. Int. Conf. Innov. Comput., Intell. Commun. Smart Electr. Syst. (ICSSES)*, Dec. 2023, pp. 1–4.
- [118] C. Islam, M. A. Babar, and A. Nepala, "Architecture-centric support for integrating security tools in a security orchestration platform," *Softw. Archit.*, vol. 3, pp. 165–181, Jan. 2020.
- [119] Z. T. Sworna, M. Ali Babar, and A. Sree Kumar, "IRP2API: Automated mapping of cyber security incident response plan to security tools' apis," in *Proc. IEEE Int. Conf. Softw. Anal., Evol. Reengineering (SANER)*, Mar. 2023, pp. 546–557.
- [120] C. Onwubiko and K. Ouazzane, *Cyber Onboarding is 'Broken'*. London, U.K.: London met repository, 2019.
- [121] M. Mutemwa, J. S. Mtsweni, and L. Zimba. (2018). *Integrating a Security Operations Center With an Organization's Existing Procedures, Policies, and Information Technology Systems*.
- [122] E. Agyepong, Y. Cherdantseva, P. Reinecke, and P. Burnap, "A systematic method for measuring the performance of a cyber security operations centre analyst," *Comput. Secur.*, vol. 124, Jan. 2023, Art. no. 102959.
- [123] J. Liu, R. Zhang, W. Liu, Y. Zhang, D. Gu, M. Tong, X. Wang, J. Xue, and H. Wang, "Context2 Vector: Accelerating security event triage via context representation learning," *Inf. Softw. Technol.*, vol. 146, Jun. 2022, Art. no. 106856.
- [124] F. D. Janos and N. H. P. Dai, "Security concerns towards security operations centers," *IEEE Xplore*, vol. 3, no. 1, pp. 000273–000278, May 2018.
- [125] G. Tsakalidis, K. Vergidis, S. Petridou, and M. Vlachopoulou, "A cyber-crime incident architecture with adaptive response policy," *Comput. Secur.*, vol. 83, pp. 22–37, Jun. 2019.
- [126] M. Baruwat Chhetri, S. Tariq, R. Singh, F. Jalalvand, C. Paris, and S. Nepal, "Towards human-AI teaming to mitigate alert fatigue in security operations centres," *ACM Trans. Internet Technol.*, vol. 24, no. 3, pp. 1–22, Aug. 2024.
- [127] R. Ganesan, S. Jajodia, and H. Cam, "Optimal scheduling of cybersecurity analysts for minimizing risk," *ACM Trans. Intell. Syst. Technol.*, vol. 8, no. 4, pp. 1–32, Jul. 2017.
- [128] M. Rosso, M. Campobasso, G. Gankhuyag, and L. Allodi, "SAIBERSOC: A methodology and tool for experimenting with security operation centers," *Digit. Threats Res. Pract.*, vol. 3, p. 10, Jun. 2021.
- [129] L. Huang and Q. Zhu, "RADAMS: Resilient and adaptive alert and attention management strategy against informational Denial-of-Service (IDoS) attacks," *Comput. Secur.*, vol. 121, Oct. 2022, Art. no. 102844.
- [130] A. Naseer, H. Naseer, A. Ahmad, S. B. Maynard, and A. Masood Siddiqui, "Real-time analytics, incident response process agility and enterprise cybersecurity performance: A contingent resource-based analysis," *Int. J. Inf. Manage.*, vol. 59, Aug. 2021, Art. no. 102334.
- [131] A. Zibak, C. Sauerwein, and A. Simpson, "A success model for cyber threat intelligence management platforms," *Comput. Secur.*, vol. 111, Dec. 2021, Art. no. 102466.
- [132] Z. C. Khan, T. Mkhwanazi, and M. Masango, "A model for cyber threat intelligence for organizations," in *Proc. Int. Conf. Artif. Intell., Big Data, Comput. Data Commun. Syst.*, 2023, pp. 1–20.
- [133] C. Daniel, T. Gill, A. R. Hevner, and M. Mullarkey, "A deep neural network approach to tracing paths in cybersecurity investigations," in *Proc. Int. Conf. Data Mining Workshops (ICDMW)*, Nov. 2020, pp. 472–479.
- [134] J. Forsberg and T. Frantti, "Technical performance metrics of a security operations center," *Comput. Secur.*, vol. 135, Dec. 2023, Art. no. 103529.
- [135] Simran, S. Kumar, and A. Hans, "The AI shield and red AI framework: Machine learning solutions for cyber threat Intelligence(CTI)," in *Proc. Int. Conf. Intell. Syst. Cybersecurity (ISCS)*, May 2024, pp. 1–6.
- [136] A. Pingle, A. Iplai, S. Mittal, A. Joshi, J. Holt, and R. Zak, "RelExt," in *Proc. IEEE/ACM Int. Conf. Adv. Social Netw. Anal. Mining*, Aug. 2019, pp. 879–886.
- [137] Z. L. Kucsán, M. Caselli, A. Peter, and A. Continella, "Inferring recovery steps from cyber threat intelligence reports," in *Lecture Notes in Computer Science*, 2024, pp. 330–349.
- [138] C. Islam, M. A. Babar, R. Croft, and H. Janicke, "SmartValidator: A framework for automatic identification and classification of cyber threat data," *J. Netw. Comput. Appl.*, vol. 202, Jun. 2022, Art. no. 103370.
- [139] V. R. Saggi, S. K. Gopal, A. S. Mohammed, S. Dhanasekaran, and M. S. Naruka, "Examine the role of generative AI in enhancing threat intelligence and cyber security measures," in *Proc. 2nd Int. Conf. Disruptive Technol. (ICDT)*, Mar. 2024, pp. 537–542.
- [140] A. Presekal, A. Stefanov, I. Semertzis, and P. Palensky, "Spatio-temporal advanced persistent threat detection and correlation for cyber-physical power systems using enhanced GC-LSTM," *IEEE Trans. Smart Grid*, vol. 2, no. 1, pp. 1–10, Apr. 2024.



MOHAMAD KHAYAT received the bachelor's degree in network and communication engineering from Al Ain University of Science and Technology and the Master of Science degree in information security from United Arab Emirates University. He has a strong educational and practical background and a deep passion for information security. His research interests include information security, the Internet of Things (IoT), artificial intelligence (AI), and machine learning (ML).

He has dedicated his academic career to exploring the intricacies of these fields and finding innovative solutions to address emerging challenges.



EZEDIN BARKA (Member, IEEE) received the Ph.D. degree in information technology from George Mason University, Fairfax, VA, USA, in 2002. He is currently an Associate Professor and the Department Chair of the College of Information Technology, United Arab Emirates University. He was a member of the Laboratory for Information Security Technology (LIST), George Mason University. His current research interests include access control, digital rights management (DRM),

large-scale security architectures and models, trust management, security in UAVs, and network "wired & wireless" and distributed systems security, where he published a number of articles addressing delegation of rights using RBAC. He has published over 50 journals and conference papers. He is a member of the IEEE Communications Society and IEEE Communications & Information Security Technical Committee (CISTC). He serves on the technical program committees for many international IEEE conferences, such as ACSAC, GLOBECOM, ICC, WIMOB, and WCNC. In addition, he has been a reviewer of several international journals and conferences.



chain, cloud federation and monitoring, non-invasive smart health monitoring, management of communities of web services, and web services applications and security. He has a large experience earned throughout his involvement and management of different research and development projects. He has served as an Organizer and the Technical Program Chair/Co-Chair for many international conferences and workshops (e.g., IWCMC'23, BDSIC'23, SERVICES'21, ICWS'20, AI-PA'2020, IIT'13, NDT'12, and WMCS'12).

MOHAMED ADEL SERHANI received the Ph.D. degree in computer engineering from Concordia University, in 2006. He is currently a Professor with the College of Computing and Informatics, University of Sharjah. He has published over 180 refereed publications, including journals, conferences, a book, and book chapters. His research interests include federated learning, cloud for data intensive e-health applications, services, SLA enforcement in cloud data centers, big data value



blockchains, smart healthcare systems, cybersecurity, and the IoT.

KHALED SHUAIB received the B.E. and M.S. degrees in electrical engineering from The City College of New York, in 1991 and 1993, respectively, and the Ph.D. degree in electrical engineering/communication networks from the Graduate Center, The City University of New York, in 1999. Since 2002, he has been with the College of Information Technology (CIT), United Arab Emirates University, where he is currently a Professor. He has over 130 refereed publications in journals and conferences and three U.S. patents. His research interests include



College of Information Technology, United Arab Emirates University (UAEU). He was with Sigpro Wireless Company, Ottawa, from August 2001 to August 2002. He joined UAEU, in September 2002, where he was a Principal Investigator (PI) and a Co-PI on several research projects. He has published over 70 refereed publications, including journals and conferences. His research interests include quality of service provisioning in wired and wireless networks, routing in wireless sensor networks, performance evaluation of wireless sensor networks, network management, modeling and simulation, the Internet of Things, and smart healthcare systems. He has served on technical program committees for many international IEEE conferences. In addition, he has been a reviewer of several international journals.

FARAG SALLABI (Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical and electronic engineering from Benghazi University, Benghazi, Libya, in 1987 and 1995, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Ottawa, Canada, in 2001. He is currently an Associate Professor with the Department of Computer and Network Engineering and the Coordinator of the Ph.D. Program of Informatics and Computing,



HEBA M. KHATER received the B.Sc. and M.Sc. degrees from the Faculty of Computers and Informatics, Department of Information Technology, Zagazig University, Egypt. Her research interests include computer networking, cyber-physical systems, security, fault-management computing, smart healthcare, and cloud computing.

...